Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

MSDW
618297

Multiple Sulfatase Deficiency, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

MUM1B 603217 MUM-1/IRF4 Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUM1 70514 MUM-1/IRF4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CMUMP Mumps Virus Antibodies, IgG and IgM, Spinal Fluid

81435

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Submit aliquot from collection vial 1.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days

	Frozen	14 days

CPT Code Information:

86735 x 2

MPPG 34947 Mumps Virus Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

MMPGM 61854 Mumps Virus Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735-Mumps, IgG

86735-Mumps, IgM

MMPM
80977

Mumps Virus Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

MUMPR 617823 Mumps Virus, Molecular Detection, PCR, Buccal

Specimen Requirements:

Specimen Type: Buccal Swab

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Sterile container with transport media

Specimen Volume: Entire collection

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosal surface around buccal cavity (the space near the upper rear molars between the cheek and the teeth) to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4, M5, Barthels FlexTrans Media or Jiangsu Transport Media)

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MUMPU 617824 Mumps Virus, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87798

MBX 70594 Muscle Pathology Consultation

Specimen Requirements:

Preferred:

Specimen Type: Frozen muscle biopsy tissue

Supplies: Muscle Biopsy Kit (T541)

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Additional Information:

1. All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Pathology accession/case number

-Anatomic site

2. Contact the Mayo Clinic Muscle Laboratory at 800-533-1710 for special problems to maximize benefit of the muscle biopsy.

Acceptable:

Specimen Type: Stained muscle biopsy slides

Collection Instructions:

1. Submit all stains performed on the case.

2. **All specimens must be labeled with anatomic site**.

Additional Information: Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 1.5 cm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	
	Ambient	

CPT Code Information:

88342 (if appropriate)

88341 (if appropriate)

88346 (if appropriate)

88350 (if appropriate)

88305 (if appropriate)

88313 (if appropriate)

88319 (if appropriate)

88314 (if appropriate)

88321 (if appropriate)

88323 (if appropriate)

88323-26 (if appropriate)

88325 (if appropriate)

FCDU9 Muscle Relaxants, Umbilical Cord Tissue (Non-Orderable)

75784

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

MUSK
64277

Muscle-Specific Kinase (MuSK) Autoantibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86366

FMUSG
57659

Mushroom IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MUSH
82626

Mushroom, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMTFG
57679

Mustard Food IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MSTD
82801

Mustard, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MUXF3
609437

MUXF3 (Cross-reactive Carbohydrate Determinant), IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86008

FBMO
75510

MVista Blastomyces Quantitative Antigen, Fluid

Specimen Requirements:

Specimen Type: Spinal fluid (CSF) or bronchoalveolar fluid

Container/Tube: Sterile, leak-proof screw cap container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL of CSF or bronchoalveolar lavage fluid.

2. Submit specimen in a sterile, leak-proof screw cap container labeled with specimen type.

3. Send refrigerate.

Additional Information:

1. Specimen type is required.

2. A separate order is required for each specimen.

3. Sputolysin, sodium hydroxide, and potassium hydroxide treatment degrade the analyte detected in the assay.

Specimen Minimum Volume: Spinal fluid: 0.8 mL; Bronchoalveolar fluid: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87449

FMVCO 57122 MVista Coccidioides Quantitative Antigen EIA

Specimen Requirements:

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 2 mL

Collection Instructions: Collect 2 mL random urine specimen. Ship specimen refrigerated in a plastic, preservative-free urine container.

Note: Sputolysin and Sodium Hydroxide are interfering substances.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	48 hours
	Frozen	

CPT Code Information:

87449

FHST 91957 MVista Histoplasma Ag Quantitative EIA

Specimen Requirements:

Specimen Type: Bronchial washing or body fluid

Container/Tube: Sterile, leak-proof screw cap container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL of bronchial washing or body fluid in a sterile, leak-proof screw cap container.

2. **Label with specimen type (required)**.

3. Send refrigerate.

Note:

1. MiraVista will test most body fluids with the following disclaimer: The reference range and other method performance specifications have not been established for this test in this type of body fluid. The test results should be integrated into the clinical context for interpretation.

2. New York approved for Urine, Serum, Plasma, spinal fluid (CSF), and bronchoalveolar lavage (BAL) specimens.

Specimen Minimum Volume: Bronchial washing/Body fluid: 0.5 mL (Note: Minimum volume does not allow for repeats.)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87385

FHSAG 90017 MVista Histoplasma Ag Quantitative, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal Fluid

Sources: CSF

Container/Tube: Sterile container

Specimen Volume: 0.8 mL

Collection Instructions: Collect 0.8 mL of spinal fluid (CSF). Ship refrigerated, 0.8 mL of spinal fluid. Send specimen in a plastic, screw-capped vial refrigerated.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

87385

MGMR 608980 Myasthenia Gravis Evaluation with Muscle-Specific Kinase (MuSK) Reflex, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	72 hours

CPT Code Information:

86041

86043 (if appropriate)

86366 (if appropriate)

MGLE 608979 Myasthenia Gravis/Lambert-Eaton Myasthenic Syndrome Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86041

86596

86043 (if appropriate)

86366 (if appropriate)

SGTF 35860 MYB (6q23) Rearrangement FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results by fluorescence in situ hybridization testing.

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 2 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

MYC
70515

MYC Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CFCTB 623130 Mycobacteria and Nocardia Culture, Cystic Fibrosis, Varies

Specimen Requirements:

Specimen Type: Respiratory fluid

Sources: Bronchoalveolar lavage fluid, bronchial washing, sputum (saliva is **not acceptable**)

Container/Tube: Sterile container

Specimen Volume: 3 to 5 mL

Collection Instructions:

1. Collect 3 separate respiratory specimens for acid-fast smears and culture in patients with clinical and chest X-ray findings compatible with tuberculosis.

2. These 3 specimens should be collected at 8 to 24-hour intervals (24 hours when possible) and should include at least 1 first-morning specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87116-Mycobacterial Culture

87015-Mycobacteria Culture, Concentration (if appropriate)

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87176-Tissue Processing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CTB 8205 Mycobacteria and Nocardia Culture, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 1.5 mL

Specimen Type: Bone marrow

Container/Tube: Sterile container, or green top (lithium or sodium heparin)

Specimen Volume: Entire collection

Specimen Type: Gastric washing

Container/Tube: Sterile container

Specimen Volume: 10 mL

Collection Instructions: Neutralize specimen within 4 hours of collection with 100 mg of sodium carbonate per 5 to 10 mL of gastric wash.

Specimen Type: Respiratory

Sources: Bronchoalveolar lavage fluid, bronchial washing, sputum

Container/Tube: Sterile container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 respiratory specimens for acid-fast smears and culture in patients with clinical and chest X-ray findings compatible with tuberculosis.

2. These 3 specimens should be collected at 8 to 24-hour intervals (24 hours when possible) and should include at least 1 **first-morning** specimen.

Specimen Type: Stool

Supplies: Stool Collection Kit, Random (T635)

Container/Tube: Sterile container

Specimen Volume: 5 to 10 g

Specimen Type: Tissue

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Collect a fresh tissue specimen.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 20 to 50 mL

Collection Instructions: Collect a random urine specimen.

Fresh tissue or body fluid are preferred over a swab specimen. Recovery of mycobacteria from swabs is generally very low yield. Only submit a swab specimen if tissue biopsy or fluid aspiration is not possible.

Specimen Type: Swab

Supplies: BD E-swab (T853)

Sources: Surgically collected wound, tissue, or body fluid

Container/Tube: Sterile, screw-capped tube containing Liquid Amies Medium with flocked swab (eg, E-swab)

Specimen Volume: Flocked swab in 1 mL of Liquid Amies Medium

Collection Instructions:

1. Before collecting specimen, wipe away any excessive amount of secretion and discharge, if appropriate.

2. Obtain secretions or fluid from source with sterile flocked swab. **Paranasal sinus**

collections must use a nasopharyngeal flocked swab.

3. Place flocked swab in sterile, screw-capped tube containing 1 mL of Liquid Amies Medium.

4. If smear and culture are requested or both a bacterial culture and mycobacterial culture are requested, collect a second swab to maximize test sensitivity. Submit each swab in a separate sterile, screw-capped tube with 1 mL of Liquid Amies Medium.

Additional Information: Swabs from the following sources are **not acceptable**: respiratory fluid (eg, sputum), nasal, sinus, outer ear, mouth, throat, or scalp. Surgically collected inner ear swabs **are** acceptable.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87116-Mycobacterial Culture

87015-Mycobacteria Culture, Concentration (if appropriate)

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87176-Tissue Processing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

CTBBL
82443

Mycobacterial Culture, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: SPS (sodium polyanethol sulfonate)

Specimen Volume: 8 to 10 mL per culture

Collection Instructions:

1. Send whole blood specimen in original tube.

2. SPS tubes are acceptable, but not preferred.

3. Note: SPS tubes must be clearly labeled as SPS. If label is obscured, sample may be canceled, as ACD (yellow top) is not an acceptable tube type.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

87116-Mycobacterial Culture

87118-Id MALDI-TOF Mass Spec AFB (if appropriate)

87150-Id, Mtb Speciation, PCR (if appropriate)

87153-Mtb PZA Confirmation, pcnA sequence (if appropriate)

87153-Mycobacteria Identification by Sequencing (if appropriate)

87150- Id, MTB complex Rapid PCR (if appropriate)

DMAID 619537 Mycobacterium abscessus Complex Subspecies Identification, Next-Generation Sequencing

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Specimen Type: *Mycobacterium abscessus* isolate growing in pure culture

Container/Tube:

Preferred: Solid slant medium (eg. Middlebrook 7H10, 7H11 or Lowenstein Jensen agar)

Acceptable: Broth medium (eg. Mycobacteria Growth Indicator Tube, 7H9 broth, BACT/ALERT MP or VersaTREK bottle). **Note:** Broth specimens will require subculture which will delay results**.**

Specimen Volume:

Solid slant medium: Isolate with visible growth

Broth medium: Greater than or equal to 3 mL of broth culture

Collection Instructions:

1. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153

TBSP 607364 Mycobacterium tuberculosis Complex Species Identification, PCR, Varies

Specimen Requirements:

Specimen Type: *Mycobacterium tuberculosis* complex isolate growing in pure culture

Supplies: Infectious Container, Large (T146)

Container/Tube: Growth on solid slant media, eg, Middlebrook 7H10, 7H11 and Lowenstein Jensen; growth in broth medium, eg, Mycobacteria Growth Indicator Tube, 7H9 broth BACT/ALERT MP or VersaTREK

Specimen Volume: Isolate with visible growth on solid media; if broth is sent, 3 mL or more of broth culture

Collection Instructions:

1. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures**.

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

Additional Information: If subculture to Middlebrook agar medium is needed to ensure purity, turnaround time for results may be delayed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87150

MTBXS 619564 Mycobacterium tuberculosis Complex, Molecular Detection and Rifampin Resistance, PCR, Sputum

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

Specimen Type: Sputum (undigested)

Container/Tube: Sterile container

Specimen Volume: 3 mL

Specimen Stability Information: Refrigerated (preferred) 7 days/Ambient 72 hours

Additional Information:

1. If a single specimen is being shared between mycobacterial culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 3 mL for the respiratory specimen is required. Specimen volumes less than indicated may decrease sensitivity of testing.

2. If insufficient volume is submitted, testing will be canceled.

Specimen Type: N-acetyl-L-cysteine/sodium hydroxide (NALC/NaOH)-digested sputum

Container/Tube: Sterile container

Specimen Volume: 3 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Stability Information: Refrigerated 7 days

Additional Information:

1. If a single specimen is being shared between mycobacterial culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 3 mL for the respiratory specimen is required. Specimen volumes less than indicated may decrease sensitivity of testing.

2. If insufficient volume is submitted, testing will be canceled.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Sputum	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

87564

DMTBR 622531 Mycobacterium tuberculosis Complex, Molecular Detection of Drug Resistance Markers, Next-Generation Sequencing

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Specimen Type: *Mycobacterium tuberculosis* complex isolate growing in pure culture

Container/Tube:

Preferred: Solid slant medium (eg. Middlebrook 7H10, 7H11 or Lowenstein Jensen agar)

Acceptable: Broth medium (eg. Mycobacteria Growth Indicator Tube, 7H9 broth, BACT/ALERT MP or VersaTREK bottle). **Note:** Broth specimens will require subculture which will delay results.

Specimen Volume:

Solid slant medium: Isolate with visible growth

Broth medium: Greater than or equal to 3 mL of broth culture

Collection Instructions:

1. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

2. Place specimen in a large infectious container and label as an etiologic agent/infectious substance.

3. Turnaround time for results may be delayed if subculture is needed to ensure purity.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153 - *M tuberculosis* drug resistance, NGS

MTBT 62203 Mycobacterium tuberculosis Complex, Molecular Detection, PCR, Paraffin, Tissue

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

Preferred:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Sources: Body tissue

Container/Tube: Tissue block

Collection Instructions: Submit a FFPE tissue block to be cut and returned.

Acceptable:

Specimen Type: Formalin-fixed, paraffin-embedded (FFPE) tissue sections (scrolls)

Sources: Body tissue

Container/Tube: Sterile containers, one for each individual cut section (scroll).

Collection Instructions:

1. Perform microtomy and prepare five separate 10-micron sections.

2. **Place each section (scroll) in a separate sterile container for submission.**

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87556

MTBRP 88807 Mycobacterium tuberculosis Complex, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit a raw clinical sample (not a culture isolate) for testing.

Fresh, undigested specimens are preferred. If no fresh specimen is available, digested respiratory specimens treated with N-acetyl-L-cysteine/sodium hydroxide (NALC/NaOH) are acceptable for some specimen types. For detailed information, see the **Specimen Type** information below.

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Mycobacterium tuberculosis* DNA is unlikely.

If a single specimen is being shared between mycobacteria culture, acid-fast smear, and/or *M tuberculosis* PCR, a minimum volume of 2 mL for body fluid, 3 mL for respiratory specimen, or a pea-sized piece of tissue should be obtained. Specimen

volumes less than indicated may decrease sensitivity of testing. If insufficient volume is submitted, test or tests will be canceled.

Submit only 1 of the following specimens:

Preferred Specimen Types

Specimen Type: Body fluid

Sources: Body, bone marrow aspirate, ocular, or CSF

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Only fresh, non-NALC/NaOH-digested body fluids are acceptable.

Specimen Type: Gastric washing

Container/Tube: Sterile container

Specimen Volume: 2 mL

Specimen Type: Respiratory

Sources: BAL, bronchial washing, tracheal secretion, or sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL if only PCR ordered or 3 mL if PCR ordered with smear and culture

Additional Information: Before collecting sputum specimens, see Ordering Guidance.

Specimen Type: Feces

Container/Tube: Sterile container

Specimen Volume: 5 to 10 g

Additional Information: Only fresh, non-NALC/NaOH-digested fecal specimens are acceptable.

Specimen Type: Tissue

Sources: Fresh tissue, bone, or bone marrow biopsy

Container/Tube: Sterile container

Specimen Volume: 5 to 10 mm

Collection Instructions: Keep moist with sterile water or sterile saline

Additional Information: Only fresh, non-NALC/NaOH-digested tissue is acceptable.

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Collect a random urine specimen.

Acceptable Specimen Types

Specimen Type: NALC/NaOH-digested respiratory specimens

Sources: Lavage fluid, bronchial washing, gastric washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid, nondigested respiratory specimen, urine: 0.5 mL; Fresh tissue or bone: 5 mm; NALC-NaOH-digested specimen, gastric washing: 1 mL; Stool: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87556-*Mycobacterium tuberculosis*, complex, molecular detection, PCR

87015-Mycobacteria culture, concentration (if appropriate)

MTBPZ 56099 Mycobacterium tuberculosis Complex, Pyrazinamide Resistance by pncA DNA Sequencing, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87153-Mtb PZA Confirmation, pncA Sequence

MPA 81563 Mycophenolic Acid, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Draw blood immediately before the next scheduled dose (trough).

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

80180

RPMPM 610248 Mycoplasma (Mycoplasmoides) pneumoniae Macrolide (Azithromycin) Resistance Prediction, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma (Mycoplasmoides) pneumoniae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-swab (T853)

-Culture Swab-Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Sources: Throat, nasal, or nasopharyngeal

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Culture transport swab (Stuart's media) or place swab in M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: Swab

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pleural, pericardial, cerebrospinal

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, tracheal secretions, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: Respiratory: 0.5 mL
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

Mycoplasma (Mycoplasmoides) pneumoniae with Macrolide Resistance Reflex, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma (Mycoplasmoides) pneumoniae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-Culture Swab-Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Sources: Throat, nasal, or nasopharyngeal

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Culture transport swab (Stuart's media) or place swab in M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: Swab

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pleural, pericardial, cerebrospinal

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Type: Respiratory

Sources: Bronchial washing, bronchoalveolar lavage, tracheal secretions, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: Respiratory specimen: 0.5 mL
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87581

MMGEN 620733 Mycoplasma genitalium, Transcription-Mediated Amplification, Post-Prostatic Massage Fluid/Urine or Peritoneal Fluids

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Post-prostatic massage fluid/urine (VBIII)

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should void a small amount of urine prior to prostatic massage. Pre-massage urine can be discarded.

3. Patient should cease voiding and a prostatic massage should be performed by urologist or other healthcare professional.

4. Collect post-massage urine into a sterile, plastic, preservative-free container.

5. Transfer 2 mL of urine into the Aptima Urine Specimen Transport Tube using the disposable pipette provided within 24 hours of collection. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

6. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

7. Transport and store post-prostatic massage fluid/urine (VBIII) specimen transport container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. **Specimen cannot be sent frozen**.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima Specimen Transfer Tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen by paracentesis using needle and syringe using routine procedures. Refer to the references below.

2. Thoroughly vortex specimen upon receipt to the laboratory.

3. Transfer 1 mL of peritoneal fluid into the Aptima Specimen Transfer kit using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug) within 24 hours of collection.

4. Recap the specimen transfer tube tightly.

5. Transport and store specimen container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. **Specimen cannot be sent frozen**.

Specimen Minimum Volume: Peritoneal fluid: 1 mL; Post-prostatic massage fluid/urine (VBIII): See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		APTIMA VIAL

CPT Code Information:

87563

AMGEN 616513 Mycoplasma genitalium, Transcription-Mediated Amplification, Urine or Urogenital Swab

Specimen Requirements:

Submit only 1 of the following specimens:

Swab specimen must be collected using an Aptima Collection Unisex Swab or Aptima Collection Multitest Swab based on the corresponding source. These swabs are contained in the Aptima Collection Kit.

Specimen Type: Endocervix/cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit**

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place the blue shaft swab into transport tube provided in collection kit.

6. Snap off swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit.**

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Urethra (Male patients only)

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit.**

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate swab gently at least 1 full rotation using sufficient pressure to ensure swab comes in contact with all urethral surfaces. Allow swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the Aptima Urine Specimen Transport Tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport container at 2 to 30 degrees C (refrigerate is preferred temperature) within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C up to an additional 90 days.

Specimen Type: Penile/meatal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit.**

2. Roll the swab (pink shaft) just at the tip or outside the opening to the penis. Be sure to roll the swab completely around the opening to get the best sample.

3. Immediately place the pink swab into the collection tube.

4. Carefully break the pink swab shaft at the score line and recap the tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate is preferred temperature) within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for an additional 90 days.

Specimen Minimum Volume: Urine-first void: 2 mL; Swabs (endocervical, urethral, vaginal, penile/meatal): See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		APTIMA VIAL

CPT Code Information:

87563

MYCO
48394

Mycoplasma pneumoniae Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738 x 2-*Mycoplasma pneumoniae* by EIA

86738-*Mycoplasma pneumoniae* by indirect IFA (if appropriate)

MYCOG 48317 Mycoplasma pneumoniae Antibodies, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma pneumoniae* Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738

MYCOM 48318 Mycoplasma pneumoniae Antibodies, IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma*

pneumoniae Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86738

MYCON 48319 Mycoplasma pneumoniae Antibody Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MYCO / *Mycoplasma pneumoniae* Antibodies, IgG and IgM, Serum.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CXCFX 601508 MYD88 Reflex to CXCR4 Mutation Detection, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see LPLFX / Lymphoplasmacytic Lymphoma/Waldenstrom Macroglobulinemia, *MYD88* L265P with Reflex to *CXCR4*, Varies.

Specimen Minimum Volume: Whole blood, Bone marrow: 1 mL; Extracted DNA: 20 mcL with a concentration of at least 10 nanograms per mcL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81479

MYD88 62927 MYD88, L265P, Somatic Gene Mutation, DNA Allele-Specific PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 50 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81305

MOGFS 65563 Myelin Oligodendrocyte Glycoprotein (MOG-IgG1) Fluorescence-Activated Cell Sorting (FACS) Assay, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection should occur prior to initiation of immunosuppressant medication.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86363

86363-titer (if appropriate)

MAGES 607034 Myelin-Associated Glycoprotein Autoantibodies, IgM, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83520

MDSDF 614276 Myelodysplastic Syndrome (MDS), Diagnostic FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 12, 88275 x 6, 88291-FISH Probe, Analysis, Interpretation; 6 probe sets

88271 x 2, 88275-FISH Probe, Analysis; each additional probe set (if appropriate)

MDSMF 614287 Myelodysplastic Syndrome (MDS), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271 x2, 88275 x1 – FISH Probe, Analysis; each additional probe set (if appropriate)

MYEFL 63414 Myelodysplastic Syndrome, Flow Cytometry, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (heparin)

Specimen Volume: 2-5 mL

Slides: Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each) x18

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

MSTF
35844

Myeloid Sarcoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be

attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete myeloid sarcoma FISH panel is ordered, include 10 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88291

88271 x 2 (if appropriate)

88271 x 2 (if appropriate)

88271 (if appropriate)

88271 x 2 (if appropriate)

88271 x 3 (if appropriate)

88274 w/modifier 52 (if appropriate)

88274 (if appropriate)

88275 (if appropriate)

MSMRD 616030 Myeloma Stratification and Risk-Adapted Therapy with Reflex to Measurable/Minimal Residual Disease, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers

MFCDF 614298 Myeloma, High Risk with Reflex Probes, Diagnostic FISH

Evaluation, Fixed Cell Pellet

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 Fixed cell pellet

Collection Instructions: Place specimen in a sterile container with a 3:1 methanol:glacial acetic acid (or similar) fixative.

Transport Temperature:

Specimen Type	Temperature	Time
Fixed Cell Pellet Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x6, 88275x3, 88291 x1-FISH Probe, Analysis, Interpretation; 3 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

MAS1 605125 Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 15

86053

86363

86341

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)
84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86363 MOGTS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86053 NMOTS (if appropriate)

84182 PC1BS (if appropriate)

86256 PC1TS (if appropriate)

86256 PC2TS (if appropriate)

86255 SP7CS (if appropriate)

86256 SP7TS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

MAC1
605126

Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 17

86053

86341

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AINCC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86255 APBCC (if appropriate)

86256 APBTC (if appropriate)

86256 CRMTC (if appropriate)
84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86053 NMOTC (if appropriate)

84182 PC1BC (if appropriate)

86256 PC1TC (if appropriate)

86256 PC2TC (if appropriate)

86255 SP7CC (if appropriate)

86256 SP7TC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

MYPO 70511 Myeloperoxidase (MPO) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MPO 80389 Myeloperoxidase Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

MPNCM 65115 Myeloproliferative Neoplasm, CALR with Reflex to MPL, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9

81339 -MPL (MPL proto-oncogene, thrombopoietin receptor) (eg, myeloproliferative disorder) gene analysis; sequence analysis, exon 10 (if appropriate)

MPNJM 606805 Myeloproliferative Neoplasm, JAK2 V617F with Reflex to CALR and MPL, Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Note: Extracted DNA from bone marrow is **not** acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81270-JAK2 (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9 (if appropriate)

81339-MPL (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence (if appropriate)

MPNR 63031 Myeloproliferative Neoplasm, JAK2 V617F with Reflex to CALR and MPL, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81270-JAK2 (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant
81219-CALR (calreticulin) (eg, myeloproliferative disorders), gene analysis, common variants in exon 9 (if appropriate)
81339-MPL (MPL proto-oncogene, thrombopoietin receptor) (eg, myeloproliferative disorder) gene analysis; sequence analysis, exon 10 (if appropriate)

GNMY9 619299 MYH9-Related Disorders, MYH9 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

MEF2B 619494 Myocyte Enhancer Factor 2 (MEF2B) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYODT 619685 MYOD1 Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88381-Microdissection, manual

81479

MYOD1 70518 Myogenic Differentiation Antigen 1 (MYOD1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYOGE 70516 Myogenin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYOGL
70517

Myoglobin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MYGLS 621090 Myoglobin, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83874

FMMPP MyoMarker 3 Plus Profile
75594

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Separate serum from cells within 1 hour of collection and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 4 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516 x 10

86235 x 7

FMYO3 MyoMarker 3 Profile
75595

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is

acceptable. Separate serum from cells within 1 hour of collection and send 5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 4 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83516 x 9

86235 x 7

MSAES 622116 Myositis Specific Antibody Evaluation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86255 x10

82397

86235

84182-SRPBS (if appropriate)

86256-SRPTS (if appropriate)

DDITF 35265 Myxoid/Round Cell Liposarcoma, 12q13 (DDIT3 or CHOP) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

NAACD
621423

N-Acetylaspartic Acid, Canavan Disease, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen (first morning void preferred).

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	90 days
	Refrigerated	90 days
	Ambient	14 days

CPT Code Information:

83921

NAT2Q
610617

N-Acetyltransferase 2 (NAT2) Genotype, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell or liver transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479-Unlisted molecular pathology procedure

NMH24 N-Methylhistamine, 24 Hour, Urine

605135

Specimen Requirements:

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot into plastic tube and send at refrigerate temperature.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

NMH1D 605159 N-Methylhistamine, 24 Hour, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NMH24 / *N*-Methylhistamine, 24 Hour, Urine.

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot into plastic tube and send at refrigerate temperature.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

NMHR
604981

N-Methylhistamine, Random, Urine

Specimen Requirements:

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

82570

NMHR1
605015

N-Methylhistamine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NMHR / *N*-Methylhistamine, Random, Urine.

Patient Preparation: Patients taking monoamine oxidase inhibitors (MAOIs) may have increased levels of *N*-methylhistamine. If medically feasible, for 2 weeks before specimen collection, patient **should not** take MAOIs.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

82542

FINA
91447

NAbFeron (IFNB-1) Neutralizing Antibody Test

Specimen Requirements:

Patient Preparation: Specimen should be collected either before treatment with interferon or more than 24 hours following the most recent dose. Patient **should not** be on steroid therapy for at least two weeks prior to testing.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Within 48 hours of collection, centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

86382

FNAD
80761

Nadolol, Serum/Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only 1 of the following specimens:

Specimen Type: Serum

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

3. Send refrigerated.

Specimen Type: Plasma

Collection Container/Tube: Lavender top or pink top (EDTA) (plasma gel/PST are **not acceptable**)

Specimen volume: 1 mL Plasma

Collection Instructions:

1. Promptly centrifuge and aliquot plasma into a plastic vial.

2. Label specimen as plasma.

3. Send refrigerated.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80299

FNALO
91784

Naloxone - Total (Conjugated/Unconjugated), Screen, Urine

Specimen Requirements:

****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect 1 mL random urine specimen in a plastic preservative-free urine container.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

80307

80362 if appropriate

G0480 if appropriate

NAPSN 70519 Napsin A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NARC 82026 Narcolepsy-Associated Antigen, HLA-DQB1 Typing, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A), lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original vial. **Do not aliquot**.

Additional Information: Specimen acceptability is based on extracted DNA concentration and not sample age.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

81376-HLA Class II typing, low resolution (eg, antigen equivalents); one locus (eg, HLA-DRB1/3/4/5, -DQB1, -DQA1, -DPB1, or -DPA1), each

QNKS 60616 Natural Killer (NK)/Natural Killer T-Cell Subsets, Quantitative, Blood

Specimen Requirements:

Container/Tube: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	30 hours	GREEN TOP/HEP

CPT Code Information:

86356 x3

86359

86357

NMS1 603542 Necrotizing Myopathy Evaluation, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge within 2 hours of collection and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

82397

86256 (if appropriate)

84182 (if appropriate)

FNECT 57941 Nectarine (Prunus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

NECT4 622259 Nectin-4, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

NEGCT 70410 Negative Control, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

GCCON 623156 Neisseria gonorrhoeae Confirmation, Nucleic Acid Amplification, Varies

Specimen Requirements:

If testing is being performed on a previously tested specimen that resulted as positive, submit that specimen.

If testing is being performed on a newly collected specimen, submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Collection Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Collection Unisex Swab. Specimens must be collected using either of these options.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection. transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerated temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Collection Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 2 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Type: Rectal/Anal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is

preferred) or -20 to -70 degrees C and transport within 14 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87591

MGRNA 61646 Neisseria gonorrhoeae, Miscellaneous Sites, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Ocular (corneal/conjunctiva)

Supplies:

-Aptima Unisex Swab Collection Kit (T583)

-Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab or Aptima Collection Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab site using Aptima Multitest Swab or Aptima Collection Unisex Swab. Specimens must be collected using either of these options.

Note: The white swab provided within the collection kit is a cleaning swab and should not be used for collection. Discard the white cleaning swab.

2. Place collection swab in transport tube provided in collection kit.

3. Snap off swab at score line so it fits into closed tube.

4. Cap tube securely and label tube with patient's entire name and collection date and

time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: Peritoneal fluid (pelvic wash, cul-de-sac fluid)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Transfer 1 mL of peritoneal fluid directly into the Aptima specimen transfer tube within 24 hours of collection.

2. Cap tube securely and label tube with patient's entire name and collection date and time.

3. Maintain Aptima specimen transfer tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

87591

GCRNA 61552 Neisseria gonorrhoeae, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimens must be collected using the Aptima Unisex Swab Collection Kit.

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

3. Discard the cleaning swab.

4. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

5. Place blue swab into transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Specimen Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely, and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urethra (Males Only)

Supplies: Swab, Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimen must be collected using the Aptima Unisex Swab Collection Kit.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely, and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection. transfer 2 mL of urine into the urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the urine transport

tube.

4. Place the labels on the transport tube so the black fill lines are still visible for volume confirmation at Mayo Clinic Laboratories.

5. Maintain urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. **ThinPrep specimen must be aliquoted (as outlined below) before it is processed for Papanicolaou testing (Pap smear).**

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima specimen transport tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima Specimen Transfer Tube tightly and gently invert 3 times to mix.

5. Label Aptima specimen transfer tube with appropriate label.

6. Use remainder of ThinPrep specimen for Pap testing.

7. Maintain specimen transport tube at 2 to 8 degrees C (refrigerate temperature is preferred) and transport within 30 days of collection, or within 14 days if stored at 15 to 30 degrees C. If longer storage is needed, freeze at -20 to -70 degrees C for 12 months.

Specimen Type: Oropharynx/Pharynx/Throat

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Collection Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Swab site using Aptima Collection Multitest Swab (pink shaft).

3. Place pink swab in transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Type: Rectal/Anal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected using Aptima Multitest Swab Specimen Collection Kit.

2. Insert swab into rectum about 3 to 5 cm past anal margin and gently rotate swab for 10 seconds.

3. Place collection swab in transport tube provided in collection kit.

4. Snap off swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL

	Ambient	APTIMA VIAL
	Frozen	APTIMA VIAL

CPT Code Information:

87591

SCGCV 621939 Neisseria gonorrhoeae, Self-Collect, Amplified RNA, Vaginal (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGV / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Vaginal.

Specimen Type: Vaginal

Supplies: Aptima Vaginal Swab Self-Collection Kit (T1001)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1001 or available separately)**.**

2. Provide patient with the Aptima Vaginal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred) and transport within 60 days of collection. If longer storage is needed, freeze at -20 to -70 degrees C for up to 12 months.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL

	Ambient	APTIMA VIAL
	Frozen	APTIMA VIAL

CPT Code Information:

SCGCR 621936 Neisseria gonorrhoeae, Self-Collect, Nucleic Acid Amplification, Rectal (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see SCCGR / *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Rectal

Specimen Type: Rectal/anal

Supplies: Aptima Rectal Swab Self-Collection Kit (T1000)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab (provided in T1000 or available separately)**.**

2. Provide patient with the Aptima Rectal Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerate temperature is preferred) or -20 to -70 degrees C and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

SCGCT 621933

Neisseria gonorrhoeae, Self-Collect, Nucleic Acid Amplification, Throat (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see: SCCGT/ *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, Self-Collect, Amplified RNA, Throat

Specimen Type: Oropharynx/pharynx/throat

Supplies: Aptima Throat Swab Self-Collection Kit (T1002)

Container/Tube: Aptima Multitest Swab

Specimen Volume: Swab

Collection Instructions:

1. **Specimens must be collected in a healthcare setting by the patient using the Aptima Multitest Swab** (provided in T1002 or available separately)**.**

2. Provide patient with the Aptima Throat Swab Self-Collection Kit or Aptima Multitest Swab and collection instructions.

3. Instruct patient to collect the specimen following the instructions provided and then return swab to the healthcare professional once complete.

4. Once patient returns the specimen, ensure the tube is securely capped, and label tube with patient's entire name and collection date and time.

5. Maintain swab container at either 4 to 30 degrees C (refrigerated temperature is preferred) or -20 to -70 degrees C, and transport within 60 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Ambient		APTIMA VIAL
	Frozen		APTIMA VIAL

CPT Code Information:

FNMEN Neisseria Meningitidis IgG Vaccine Response

91669

Specimen Requirements:

Container/Tube:

Preferred: Red top tube

Acceptable: Serum gel tube

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Note: Serum gel tube is acceptable, but must pour off into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86317/x4

FNNAT 75929 Neonatal Alloimmune Thrombocytopenia (NAIT)-Serial Monitoring Maternal only

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instruction:

1.Centrifuge and aliquot 5 mL serum into a plastic vial.

2.Send refrigerated.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

86022

86023 x 2

NBILI
82133

Neonatal Bilirubin, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: 2 Serum gel Microtainers

Acceptable: 2 Red top Microtainers

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel microtainers should be centrifuged within 2 hours of collection.

2. Red-top microtainers should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	24 hours	LIGHT PROTECTED
	Frozen	30 days	LIGHT PROTECTED
	Ambient	6 hours	LIGHT PROTECTED

CPT Code Information:

82247

82248

FNEOS
75451

Neopterin

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 0.8 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.8 mL of serum into an amber vial to protect from light.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL; Note: This volume does not allow for repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen (preferred)	180 days	LIGHT PROTECTED
	Refrigerated	72 hours	LIGHT PROTECTED

CPT Code Information:

83520

RSCGP
618128

Nephrocalcinosis, Nephrolithiasis, and Renal Electrolyte Imbalance Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404 x 4

81405 x 2

81406 x 8

81407 x 2

81479

81479 (if appropriate for government payers)

NETT
82734

Nettle, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

NEUN 70631 Neu-N Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NONCP 603047 Neuro-Oncology Expanded Gene Panel with Rearrangement, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-If ordered in conjunction with CMAPT / Chromosomal Microarray, Tumor, Formalin-Fixed Paraffin-Embedded, the preferred amount of tissue is 430 mm(2), the minimum amount is 180 mm(2).

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 6 mm x 6 mm x 10 slides as preferred: approximate/equivalent to 360 mm(2) and the minimum as 4 mm x 4 mm x 10 slides: approximate/equivalent to 144 mm(2).

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens. Tissue blocks are preferred over tissue slides.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 15

slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81455

81456

NONCM 622295 Neuro-Oncology Gene Panel, Mutations Only, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 288 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-If ordered in conjunction with CMAPT / Chromosomal Microarray, Tumor, Formalin-Fixed Paraffin-Embedded, the preferred amount of tissue is 430 mm(2), the minimum amount is 180 mm(2).

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

- For this test, at least 6 mm x 6 mm areas on 8 unstained slides is preferred: this is approximately equivalent to 288 mm(2). The minimum acceptable area is 6 mm x 6 mm on 1 unstained slides: approximately equivalent to 36 mm(2). For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing.

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Acceptable:

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the following slides:

1 Slide stained with hematoxylin and eosin

AND

15 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 15 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

Tumor

Specimen Requirements:

This assay requires at least 60% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 144 mm(2)

-These amounts are cumulative over up to 20 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 6 mm x 6 mm x 10 slides as preferred: approximate/equivalent to 360 mm(2) and the minimum as 4 mm x 4 mm x 10 slides: approximate/equivalent to 144 mm(2).

-If ordered in conjunction with other tests, please refer to specimen requirements for the specific tests.

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with an acceptable amount of tumor tissue

Acceptable:

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the following slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Unused unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81524

NCDCS 615865 Neurochondrin Antibody, Cell-Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NCDCC 615864 Neurochondrin Antibody, Cell-Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NCDTS 616111 Neurochondrin Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

NCDTC 616112 Neurochondrin Antibody, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

NF4FS 614591 Neurofascin-155 IgG4, Flow Cytometry, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-CIDP / Chronic Inflammatory Demyelinating Polyradiculoneuropathy/Nodopathy Evaluation, Serum

-DMNES / Peripheral Nervous System Demyelinating Neuropathy, Autoimmune Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

NF1Z 614585 Neurofibromatosis Type 1, NF1, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes

prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81408

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

NF2F 70520 Neurofilament (2F11) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NFSMI 71354 Neurofilament (SMI31) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NFLP 622765 Neurofilament Light Chain, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Submission Container/Tube: Plastic screw-top vial

Specimen Volume: 0.6 mL

Collection Information: Centrifuge and aliquot plasma into a plastic vial. **Do not submit specimen in original tube.**

Specimen Minimum Volume: 0.50 mL

Transport Temperature:

Specimen Type	Temperature	Time
EDTA Plasma	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

83884

PNEFS Neuroimmunology Antibody Follow-up, Serum

84300

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 AAGFS (if appropriate)

86043 ACMFS (if appropriate)

84182 AGNBS (if appropriate)

86255 AK5CS (if appropriate)

86255 AK5IS (if appropriate)

86255 AINCS (if appropriate)

86255 AMPCS (if appropriate)

86255 AMPHS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

84182 AN2BS (if appropriate)

86255 AGN1S (if appropriate)

86255 ANN1S (if appropriate)

86255 ANN2S (if appropriate)

86255 ANN3S (if appropriate)

86255 APBCS (if appropriate)

86255 APBIS (if appropriate)

86255 CS2CS (if appropriate)

86255 CRMS (if appropriate)

86255 DPPCS (if appropriate)

86255 DPPIS (if appropriate)

86255 GABCS (if appropriate)

86255 GFACS (if appropriate)

86255 GFAIS (if appropriate)

86255 GRFCS (if appropriate)

86255 GRFIS (if appropriate)

83516 IGG_D (if appropriate)

86255 IG5CS (if appropriate)

86255 IG5IS (if appropriate)

83516 IGM_D (if appropriate)

83516 IGM_M (if appropriate)

86255 ITPCS (if appropriate)

86255 ITPIS (if appropriate)

86255 LG1CS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86255 GL1IS (if appropriate)

86255 GL5CS (if appropriate)

86255 GL5IS (if appropriate)

86255 NCDCS (if appropriate)

86255 NCDIS (if appropriate)

86255 NFHCS (if appropriate)

86255 NIFIS (if appropriate)

86255 NFLCS (if appropriate)

86255 NMDCS (if appropriate)

83519 CCPQ (if appropriate)

84182 PC1BS (if appropriate)

84182 PCTBS (if appropriate)

86255 PCABP (if appropriate)

86255 PCAB2 (if appropriate)

86255 PCATR (if appropriate)

86255 PDEIS (if appropriate)

86255 SP5CS (if appropriate)

86255 SP5IS (if appropriate)

86255 SP7CS (if appropriate)

86255 SP7IS (if appropriate)

86255 SRPIS (if appropriate)

84182 SRPBS (if appropriate)

86255 T46CS (if appropriate)

86255 T46IS (if appropriate

PNEFC Neuroimmunology Antibody Follow-up, Spinal Fluid

84299

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Vial number 1

Acceptable: Any vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 AK5CC (if appropriate)

86255 AK5IC (if appropriate)

84182 AGNBC (if appropriate)

86255 AINCC (if appropriate)

86255 AMPCC (if appropriate)

86255 AMPHC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

84182 AN2BC (if appropriate)

86255 AGN1C (if appropriate)

86255 ANN1C (if appropriate)

86255 ANN2C (if appropriate)

86255 ANN3C (if appropriate)

86255 APBCC (if appropriate)

86255 APBIC (if appropriate)

86255 CS2CC (if appropriate)

84182 CRMWC (if appropriate)

86255 CRMC (if appropriate)

86255 DPPCC (if appropriate)

86255 DPPIC (if appropriate)

86255 GABCC (if appropriate)

86255 GFACC (if appropriate)

86255 GFAIC (if appropriate)

86255 GRFCC (if appropriate)

86255 GRFIC (if appropriate)

86255 IG5CC (if appropriate)

86255 IG5IC (if appropriate)

86255 ITPCC (if appropriate)

86255 ITPIC (if appropriate)

86255 LG1CC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86255 GL1IC (if appropriate)

86255 GL5CC (if appropriate)

86255 GL5IC (if appropriate)

86255 NCDCC (if appropriate)

86255 NCDIC (if appropriate)

86255 NFHCC (if appropriate)

86255 NIFIC (if appropriate)

86255 NFLCC (if appropriate)

86255 NMDCC (if appropriate)

84182 PC1BC (if appropriate)

84182 PCTBC (if appropriate)

86255 PCTRC (if appropriate)

86255 PCA1C (if appropriate)

86255 PCA2C (if appropriate)

86255 PDEIC (if appropriate)

86255 SP5CC (if appropriate)

86255 SP5IC (if appropriate)

86255 SP7CC (if appropriate)

86255 SP7IC (if appropriate)

86255 T46CC (if appropriate)

86255 T46IC (if appropriate)

NMOFS 38324 Neuromyelitis Optica (NMO)/Aquaporin-4-IgG Fluorescence-Activated Cell Sorting (FACS) Assay, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86053

86053-titer (if appropriate)

NMOFC 38325 Neuromyelitis Optica (NMO)/Aquaporin-4-IgG Fluorescence-Activated Cell Sorting (FACS) Assay, Spinal Fluid

Specimen Requirements:

Collection Container/Tube: Sterile vial

Specimen Volume: 3 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86053

86053-titer (if appropriate)

NSESF
81796

Neuron-Specific Enolase (NSE), Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	15 days
	Ambient	72 hours

CPT Code Information:

83520

NSEI
70630

Neuron-Specific Enolase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NSE
80913

Neuron-Specific Enolase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. **Specimens should not be transported by tube system prior to centrifugation**.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Ambient	5 days

CPT Code Information:

83520

NCLGP
608014

Neuronal Ceroid Lipofuscinosis (Batten Disease) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

NCLBS 616838 Neuronal Ceroid Lipofuscinosis, Two-Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete, ie, unpunched.

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood Spots: 1; Whole Blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

NCLW 616839 Neuronal Ceroid Lipofuscinosis, Two-Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

NEUD1 622267 Neuronal Differentiation 1 (NeuroD1), Technical Component only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

NSAIP 621136 Neurosyphilis IgG Antibody Index with VDRL, Serum and Spinal Fluid

Specimen Requirements:

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within a maximum of 24 hours of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. The spinal fluid (CSF) specimen **must be** collected within 24 hours of the serum specimen, preferably at the same time.

2. The CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture. **Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection**.

3. Label vial as spinal fluid or CSF.

4. Band CSF specimen together with the serum sample.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL Serum

Collection Instructions:

1. Within 24 hours of collection of the spinal fluid specimen, a serum specimen **must**

also be collected, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label tube as serum.

4. Band serum specimen together with the CSF sample.

Specimen Minimum Volume: Spinal fluid/Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	10 days
	Frozen	10 days
Serum	Refrigerated (preferred)	10 days
	Frozen	10 days

CPT Code Information:

86780

82784 x 2-(if appropriate)

82040-(if appropriate)

86780 x 2-(if appropriate)

82042-(if appropriate))

86593-(if appropriate)

NSAI 621141 Neurosyphilis IgG, Antibody Index, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NSAIP / Neurosyphilis IgG Antibody Index with VDRL, Serum and Spinal Fluid.

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within a maximum of 24 hours of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. The CSF specimen **must be** collected within 24 hours of the serum specimen, preferably at the same time.

2. The CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture. **Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection**.

3. Label vial as spinal fluid or CSF.

4. Band CSF specimen together with the serum sample.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL Serum

Collection Instructions:

1. Within 24 hours of collecting the spinal fluid specimen, a serum specimen **must also be** collected, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label tube as serum.

4. Band serum specimen together with the CSF sample.

Specimen Minimum Volume: Spinal fluid/Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	10 days
	Frozen	10 days

CPT Code Information:

82784 x 2

82040

86780 x 2

82042

FNEU2 Neurotensin
75905

Specimen Requirements:

Patient preparation:

1. Fasting: 10 to 12 hours

2. Patient should not be on any antacid medication or medications that affect gastroentero-intestinal function, if possible, for at least 48 hours prior to collection.

Supplies: Z-Tube Preservative Inter Science Institute (T701)

Collection Container/Tube: Z-tube

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Draw 10 mL of blood directly into special Z-tube

2. As soon as possible, centrifuge and aliquot plasma into a plastic vial.

3. Freeze immediately after separation.

4. Ship frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Z-Tube Plasma	Frozen	180 days

CPT Code Information:

83519

NADF
35312

Newborn Aneuploidy Detection, FISH, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

NIU
8626

Nickel, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic aliquot container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 0.55 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83885

NIS
8622

Nickel, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

Metal Free Specimen Vial (T173)

Collection Container/Tube: Royal blue-top (metal-free, no additive)

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, vial

Specimen Volume: 0.55 mL Serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a Mayo metal-free, screw-capped vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	METAL FREE
	Ambient	7 days	METAL FREE
	Frozen	7 days	METAL FREE

CPT Code Information:

83885

NIUCR
614552

Nickel/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 2.55 mL

Collection Instructions:

1. Collect a random urine specimen

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83885

82570

NIUC
614553

Nickel/Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see NIUCR / Nickel/Creatinine Ratio, Random, Urine.

Patient Preparation:

High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 2.55 mL

Collection Instructions:

1. Collect a random urine specimen

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

NICOU
82510

Nicotine and Metabolites, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

G0480

80323 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

NICOS
82509

Nicotine and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80323 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

NITU
607705

Nitrogen, Total, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative. Refrigerated is the preferred preservation method.

Specimen Stability Information: Frozen 3 years

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	

CPT Code Information:

84999

NITF
607704

Nitrogen, Total, Feces

Specimen Requirements:

Patient Preparation: Laxatives and enemas should not be used during collection as barium and boric acid interfere with test procedure.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool container; complies with shipping requirements, do not use other containers.

Specimen Volume: Entire collection (24, 48, 72, or 96 hours)

Collection Instructions:

1. All containers must be sent together.

2. Entire collection must contain at least 5 g of feces.

3. The number of containers sent should be indicated on the labels (ie, 1 of 4)

Specimen Stability Information: Frozen 3 years

Additional Information: Patient can store sample at refrigerate temperature during collection period.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84999

NKX3 606692 NKX3.1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SSF1 87294 Nocardia Stain, Varies

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 0.2 mL

Collection Instructions: Collect a raw specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87206

87176-Tissue processing (if appropriate)

NDSPC 113339 Non-Gynecologic Direct Smear (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88104

NTPPC 113337 Non-Gynecologic, ThinPrep (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88112

NSIP
31769

Non-Seasonal Inhalant Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

NSFIB
604200

Nonalcoholic Steatohepatitis (NASH)-FibroTest, Serum and Plasma

Specimen Requirements:

Both serum and plasma are required for this test.

Patient Preparation:

Fasting:12 hours, required

Specimen Type: Serum

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 4 mL

Collection Instructions:

1. Centrifuge and aliquot serum into an amber vial within 2 hours of collection.

2. Centrifuged serum must be light protected within 4 hours of collection. It is acceptable to draw the blood and then protect it from light after centrifugation as long as it's within 4 hours of collection.

3. Label specimen as serum.

Specimen Type: Plasma

Collection Container/Tube: Gray top (potassium oxalate/sodium fluoride)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Label specimen as plasma.

Specimen Minimum Volume: Serum: 2 mL
Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma NaFl-KOx	Refrigerated (preferred)	7 days	
	Frozen	14 days	

	Ambient	7 days	
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Ambient	24 hours	LIGHT PROTECTED

CPT Code Information:

0003M

NSRGG 617393 Noonan Syndrome and Related Conditions Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to

determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81442

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

LNORO 65170 Norovirus PCR, Molecular Detection, Feces

Specimen Requirements:

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium - either Cary-Blair, Para-Pak Culture and Sensitivity Media)

Specimen Volume: Representative portion of diarrheal fecal sample, 1 gram or 5 mL

Collection Instructions:

1. Collect fresh feces and place in preservative within 1 hour of collection.

2. Visibly formed feces are not consistent with Norovirus gastrointestinal disease and should not be submitted for testing.

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient	7 days

CPT Code Information:

87798 x 2

NEREG
31767

Northeast Regional Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

NOTRP
37119

Nortriptyline, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

NTC3Z 616563 NOTCH3 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese

Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FCDU4 75779 Novel Psychoactives, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

NR4A3
64940

NR4A3 (9q22.33) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be

rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-NA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PBNP1 NT-Pro B-Type Natriuretic Peptide, Serum
615897

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot the serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days

CPT Code Information:

83880

NTRKM 619703 NTRK Genes Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

NMRLP 603839 Nuclear Magnetic Resonance Lipoprotein Profile, Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 12 hours**, **required**; On night before examination, evening meal should be eaten before 6 p.m. and should contain no fatty foods.

2. Patient must not consume any alcohol for 24 hours before the specimen is collected.

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Allow isopropyl alcohol (from phlebotomy site prep) to dry thoroughly before venipuncture.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83704

NMITO 617090 Nuclear Mitochondrial Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81440

NPMFM 620654 Nucleophosmin (NPM1) Mutation Analysis, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Refrigerated (preferred)	72 hours
	Ambient	72 hours

CPT Code Information:

81310

NPM1Q Nucleophosmin (NPM1) Mutation Analysis, Varies

604418

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix whole blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	72 hours

CPT Code Information:

81310-*NPM1* (nucleophosmin) (eg, acute myeloid leukemia) gene analysis; exon 12 variants

NUT
70521

NUT Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NUT1F
63431

NUTM1 (15q14) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Container/Tube: Formalin-fixed, paraffin-embedded tumor tissue block

Specimen Type: Slides

Specimen Volume: 4 Consecutive, unstained, 5 micron-thick sections placed on positively charged slides and 1 hematoxylin and eosin-stained slide

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 2 consecutive, unstained, 5 micron- thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274-w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

NMEG
82497

Nutmeg, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

NUTSP
31771

Nuts Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

FNGPG
57930

Nuts and Grains Panel IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 17

FOAKE
57999

Oak Live (Quercus virginiana) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FROE
57907

Oak Red (Quercus rubra) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

OAK
82673

Oak, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FOATG
57576

Oat IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

OATS
82688

Oat, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OCT2
70522

OCT-2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OCT4
70523

OCT3/4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OCTO
82820

Octopus, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLNZ
91129

Olanzapine (Zyprexa)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.50 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

OLIG2 OLIG2 Immunostain, Technical Component Only
71357

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OLIGS Oligoclonal Banding, Serum (Non-Orderable)
2783

Specimen Requirements:

Only orderable as part of a profile. For more information see:

OLIG / Oligoclonal Banding, Serum and Spinal Fluid

MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916

OLIG
8017

Oligoclonal Banding, Serum and Spinal Fluid

Specimen Requirements:

Both serum and spinal fluid are required. Spinal fluid must be obtained within 7 days of serum collection.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as spinal fluid.

Specimen Minimum Volume: Serum, Spinal fluid: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916 x 2

OLIGC
3484

Oligoclonal Banding, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information, see:

OLIG / Oligoclonal Banding, Serum and Spinal Fluid

MSP3 / Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

83916

OLIGU
64889

Oligosaccharide Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 8 mL

Pediatric Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No added preservative.

3. Immediately freeze specimen.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	365 days
	Refrigerated	15 days
	Ambient	7 days

CPT Code Information:

84377

FOLBG
57671

Olive Black IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days

	Ambient	7 days

CPT Code Information:

86001

OLIV
82733

Olive Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OLIVF
86306

Olive-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FONG
57636

Onion IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ONIN
82806

Onion, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OPTMX
62736

Opiate Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653) includes the specimen containers, seals, and documentation required.

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Additional Information:

1. Specimen that arrives with a broken seal does not meet the chain of custody requirements.

2. The laboratory recommends sending chain-of-custody specimens by overnight shipment.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATM
84326

Opiate Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATS
623435

Opiate Panel, Free (Unconjugated), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80356 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATX
62735

Opiates Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 5 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 5 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80362 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OPATU 8473 Opiates Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80361 (if appropriate for select payers)

80365 (if appropriate for select payers)

80362 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDU1 Opiates, Umbilical Cord Tissue (Non-Orderable)
75776

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

FORNG Orange IgG
57632

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

ORNG Orange, IgE, Serum
82740

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ORCH
82907

Orchard Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

OREG
82496

Oregano, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ORXNA
604230

Orexin-A/Hypocretin-1, Spinal Fluid

Specimen Requirements:

Patient Preparation: Patient **should not** have recently received radioisotopes, either therapeutically or diagnostically, due to potential assay interference.

Collection Container/Tube: Sterile vial

Submission Container/Tube: Plain vial with no additives

Specimen Volume: 1.5 mL

Pediatric Volume: 0.5 mL

Collection Instructions:

1. Obtain aliquot from second collection vial (preferred, not required).

2. Hemolyzed specimens will give false-positive results. Specimens should be centrifuged to remove any red cells prior to shipping.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	120 days
	Refrigerated	7 days
	Ambient	72 hours

CPT Code Information:

83519

OAU
80619

Organic Acids Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Pediatric: If insufficient collection volume, submit as much specimen as possible in a single container; the laboratory will determine if volume is sufficient for testing.

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	416 days
	Refrigerated	14 days

CPT Code Information:

83919

OAUS
610707

Organic Acids Screen, Urine Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell filter paper

Specimen Volume: 1 Filter paper card soaked with urine, typically requires 2 to 3 mL of urine

Collection Instructions

1. Soak a filter paper card with urine, approximately 22 cm(2) in area (typically requires 2 to 3 mL of urine; exact measurement is not important as urine volume will be normalized to creatinine).

Note: Filter paper sample may be collected by dipping the card into a collection cup of urine. Avoid dilute urine if possible.

2. Let urine dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen (preferred)	28 days	FILTER PAPER
	Ambient	14 days	FILTER PAPER

	Refrigerated	14 days	FILTER PAPER

CPT Code Information:

83919

IDENT 9221 Organism Referred for Identification, Aerobic Bacteria

Specimen Requirements:

Preferred:

Specimen Type: Bacterial isolate swab

Supplies:

E-swab (T853)

Infectious Container, Large (T146)

Container/Tube:

Preferred: E-Swab collection and transport system

Acceptable: Flocked swab and 1-mL liquid Amies transport medium in 12 x 80 mm tube

Collection Instruction:

1. Perform isolation of infecting bacteria.

2. Utilize the flocked swab to obtain an adequate sample of pure cultured isolate. **Do not submit mixed cultures.**

3. Place swab into the transport system containing 1-mL liquid Amies transport medium.

4. If needed, break off end of swab and close the transport tube.

5. Place the transport system into the secondary infectious container for shipment.

6. Each isolate must be submitted under a separate order.

Note: For the following organisms, submit an agar slant or other appropriate media to ensure viability upon arrival to the laboratory; *Neisseria gonorrhoeae*, *Campylobacter* sp., *Helicobacter pylori* and any other fastidious organism.

Acceptable:

Specimen Type: Pure culture of organism from source cultured

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Collection Instructions:

1. Perform isolation of infecting bacteria.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

3. Place the agar slant or other appropriate media into the secondary infectious container for shipment.

4. **Each isolate must be submitted under a separate order.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87077-Organism Referred for Identification, Aerobic Bacteria

87077-Identification Commercial Kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 3-Serologic Agglut Method 1 Ident (if appropriate)

87147-Serologic Agglut Method 2 Ident (if appropriate)

87147 x 4-Serologic Agglut Method 3 Ident (if appropriate)

87147 x 2-6 - Serologic Agglut Method 4 Ident (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87798-Identification by PCR (if appropriate)

ANIDE
8114

Organism Referred for Identification, Anaerobic Bacteria

Specimen Requirements:

Specimen Type: Pure culture of organism from a source not normally colonized by anaerobes

Acceptable Sources: Abscesses, percutaneous transtracheal aspirates, sterile body fluids, suprapubic aspirations, or wounds

Supplies:

-Anaerobic Transport Tube (T588)

-Infectious Container, Large (T146)

Container/Tube:

Preferred: Anaerobic transport tube

Acceptable: Thioglycollate broth or any other suitable anaerobic transport system

Collection Instructions:

1. Perform isolation of infecting bacteria.

2. Bacterial organism must be submitted in pure culture, actively growing. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87076-Organism ref for ID, anaerobic bact

87076-Id MALDI-TOF mass spec anaerobe (if appropriate)

87076-Anaerobe Ident (if appropriate)

87153-Anaerobe ident by sequencing (if appropriate)

87077-Identification commercial kit (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

87077-Bacteria identification (if appropriate)

87077-Additional identification procedure (if appropriate)

87077-Identification Staphylococcus (if appropriate)

87077-Identification Streptococcus (if appropriate)

87147 x 3-Serologic agglut method 1 ident (if appropriate)

87147-Serologic agglut method 2 ident (if appropriate)

87147 x 4-Serologic agglut method 3 ident (if appropriate)

87147 x 2-6-Serologic Agglut Method 4 Ident (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87798-Identification by PCR (if appropriate)

OROT 8905 Orotic Acid, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

83921

Orris Root (Iris florentina) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

OTP
620710

Orthopedia Homeobox Protein (OTP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

OPTU
614360

Orthostatic Protein, Timed Collection, Urine

Specimen Requirements:

Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Supplies: 2 Sarstedt 5 mL Aliquot Tube (T914)

Daytime Collection

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a 16-hour (daytime) urine specimen.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not over fill aliquot tube 4 mL at most.

5. Collect specimen per instructions in Orthostatic Protein Measurement 24-Hour Urine:

Collection Site Instructions (T546) in Special Instructions.

Nighttime (Supine) Collection

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect an 8-hour (nighttime) urine specimen.

2. No preservative.

3. Invert well before taking 4-mL aliquot at most.

Specimen Minimum Volume: 1 mL from 16-hour (daytime) urine collection/1 mL from 8-hour (nighttime) urine collection

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156 x 2

ORTHP 617766 Orthostatic Proteinuria, Random, Urine

Specimen Requirements:

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. This is a nighttime (supine) collection.

2. Void immediately prior to lying down to sleep.

3. Collect an 8-hour (nighttime) urine specimen.

4. Upon awaking in the morning, the patient should immediately void into the specimen cup.

4. No preservative.

5. Invert well before taking 4-mL (maximum) aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

CRETR- 82570

PRCON- 84156

OSM24 Osmolality, 24 Hour, Urine

622326

Specimen Requirements:

Supplies:

-Diazolidinyl Urea (Germall), 5.0 mL (T822)

-Urine Tubes, 10 mL (T068)

Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours

2. Add 5 mL of diazolidinyl urea as preservative at start of collection or refrigerate specimen during and after collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83935

UOSMU 606520 Osmolality, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions: Collect a random urine specimen

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83935

UOSMS 614057 Osmolality, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

83930

FRAG 9064 Osmotic Fragility, Erythrocytes

Specimen Requirements:

Both patient and shipping control whole blood specimens are required. For complete instructions, refer to Specimen Collection and Labeling Instructions for Osmotic Fragility Testing of Erythrocytes.

Shipping Control Specimen

Specimen Type: Whole blood (non-patient)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a shipping control specimen from a healthy person unrelated to the patient at the same time (or within 4 hours) as the patient specimen.

Note: The shipping control specimen can be collected from a phlebotomist, volunteer or another heathy patient.

2. Clearly handwrite "CONTROL" on the outermost label of the tube.

3. Immediately refrigerate (or place on cold gel pack/small amount of wet ice) after collection.

4. Send shipping control in the original tube. **Do not aliquot**.

5. Keep the shipping control and the patient specimens **together**, either rubber banded or in a bag.

Additional Information:

1. **The shipping control and patient specimens must be handled identically** from the time of collection through receipt in the testing laboratory.

2. **If the shipping control is not sent with the patient specimen, test cancellation is likely.**

3. The shipping control specimen evaluates whether the patient result has been compromised by handling conditions such as temperature, motion, or other transportation interferences, as temperature and handling extremes can adversely impact the integrity of the specimen.

Patient Specimen

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect and label patient specimen.

2. Immediately refrigerate (or place on cold gel pack/small amount of wet ice) specimen after collection.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

4. Keep the shipping control and the patient specimens together, either rubber banded or in a bag.

Additional Information: The shipping control and patient specimens must be handled identically from the time of collection through receipt in the testing laboratory.

Specimen Minimum Volume: Patient whole blood, shipping control: 2 mL each

Transport Temperature:

Specimen Type	Temperature	Time	
Control	Refrigerated	72 hours	PURPLE OR PINK TOP/EDTA
Whole Blood EDTA	Refrigerated	72 hours	

CPT Code Information:

85557

OSCAL 80579 Osteocalcin, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

83937

OIBFG 617407 Osteogenesis Imperfecta and Bone Fragility Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Molecular Testing, Chorionic Villi/Products of Conception.

2. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Blood: 1 mL; Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x2

81408 x2

81479

81479 (if appropriate for government payers)

Ova and Parasite, Microscopy, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Bile

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) and/or slides

Specimen Volume: 4 mL EDTA and/or 1 slide (unstained preferred)

Collection Instructions:

1. Bone marrow and/or slides will be accepted for this test.

2. If submitting slides with EDTA tube, label and bag specimens together. Send refrigerate as one collection.

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Colonic washing

Container/Tube:

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND zinc polyvinyl alcohol (Zn PVA) (one vial of each)

Specimen Volume: at least 2 mL

Collection Instructions:

1. Place specimen into preservative within 30 minutes of collection.

2. Follow instructions on the container as follows:

a. Colonic washings should be placed into ECOFIX preservative vial (or 10% formalin

and Zn-PVA vials) in a ratio of 1 part preservative:1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

b. Place washings into vial, twist the cap tightly closed, and shake vigorously until the contents are well mixed.

c. Do not fill above the fill line indicated on the preservative container.

Specimen Stability Information: Ambient (preferred)/Refrigerate 5 days

Specimen Type: Duodenal aspirate

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND Zn PVA (one vial of each)

Specimen Volume: at least 2 mL

Collection Instructions:

1. Place specimen into preservative within 30 minutes of collection.

2. Follow instructions on the container as follows:

a. Duodenal aspirate should be placed into ECOFIX preservative vial (or 10% formalin and Zn-PVA vials) in a ratio of 1 part preservative:1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

b. Place washings into vial, twist the cap tightly closed, and shake vigorously until the contents are well mixed.

c. Do not fill above the fill line indicated on the preservative container.

Specimen Stability Information: Ambient (preferred)/Refrigerate 5 days

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: At least 1 mL

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Fluid, abscess, drainage material

Sources: Abdominal, ascites, brain, cyst (must specify anatomical location of cyst), liver, lymphatic, peritoneal, splenic

Container/Tube: Sterile container AND either ECOFIX preservative vial or one zinc (Zn)-PVA AND one 10% formalin vial

Specimen Volume: 15 mL

Collection Instructions:

1. Place half of collection into preservative (either ECOFIX or PVA and 10 % Formalin combo) in a ratio of 1 part preservative: 1 part specimen. Some preservative may need to be removed from the vial to maintain this ratio.

2. Label the preservative type on the side of each vial.

3. Place other half of collection in a sterile container.

3. Label both specimens, bag together, and send refrigerate as one collection.

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Respiratory

Source: Bronchial washing, bronchoalveolar lavage, sputum

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Stability Information: Refrigerate 5 days

Specimen Type: Tissue

Sources: Bladder, brain, colon, intestine, liver, lymph node, lung, muscle, rectal, spleen (must specify anatomical location of tissue)

Container/Tube: Sterile container

Specimen Volume: 5 to10 mm

Collection Instructions: Place specimen in sterile container with 1 to 2 drops of sterile saline to keep tissue moist.

Specimen Stability Information: Refrigerate 5 days

Specimen Minimum Volume: Respiratory specimens, spinal fluid, abscess, or drainage material: 0.5 mL
Tissue: 3 mm
Other specimen types: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	5 days

CPT Code Information:

87015-Concentration (any type), for infectious agents (if appropriate)

87209-Smear, primary source with interpretation; complex special stain (eg, trichrome, iron hematoxylin) for ova and parasites (If appropriate)

87210-Wet mount for infectious agents (if appropriate)

87207-Smear, primary source, with interpretation; special stain for inclusion bodies or intracellular parasites (if appropriate)

OPE 619736 Ova and Parasite, Travel History or Immunocompromised, Feces

Specimen Requirements:

Patient Preparation: Specimen collection should be delayed for 7 to 10 days after administration of barium, bismuth, kaolin, magnesia, castor oil or mineral oil, and 2 to 3 weeks after antibiotics have been given since these may interfere with identification of protozoa.

Specimen Type: Preserved feces (Unpreserved specimens are **not acceptable)**

Supplies: ECOFIX Stool Transport Vial (Kit) (T790)

Container/Tube:

Preferred: ECOFIX preservative

Acceptable: 10% Buffered formalin AND zinc polyvinyl alcohol (Zn PVA) (one vial of each)

Specimen Volume: Portion of stool: 2 g or 2 mL

Collection Instructions:

1. Follow instructions on the container as follows:

a. Place specimen into Ecofix preservative vial or one 10% formalin AND one Zn PVA preservative vial within 30 minutes of passage or collection. Refer to the fill line on the preservative vial. Do not fill above the line indicated on the container.

b. Mix the contents of the tube with the spoon, twist the cap to close tightly, and shake vigorously until the contents are well mixed.

Additional Information:

1. Stool placed in 10% buffered formalin can be accepted if accompanied by Zn PVA-preserved specimen; **10% buffered formalin-preserved specimens submitted without an accompanying Zn PVA-preserved specimen will be canceled**.

2. Specimen submitted in low viscosity (LV)-PVA, mercury PVA or copper (Cu)-PVA **will be rejected**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	21 days
	Refrigerated	21 days

CPT Code Information:

87177-Concentration (any type), for infectious agents

87209-Smear, primary source with interpretation; complex special stain (eg, trichrome, iron hematoxylin) for ova and parasites

OVAL
82826

Ovalbumin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

FOVAS 57836 Ovarian Antibody Screen with Reflex to Titer, IFA

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86255

86256 (if appropriate)

OVAPA 623946 Ovarian Malignancy Risk Assessment, Ova1Plus, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Information:

1. Allow the specimen to clot for at least 30 minutes.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial or send centrifuged serum gel tube.

4. For red top tubes, aliquot serum into a plastic vial within 10 minutes of centrifugation.

Specimen Minimum Volume: Serum: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days
	Frozen	90 days

CPT Code Information:
81503

0003U

OVAWA 623945 Ovarian Mass Risk Monitoring, OvaWatch, Serum

Specimen Requirements:

Collection Container/Tube: Sarstedt Aliquot Tube, 5 mL (T914)

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Information:

1. Allow the specimen to clot for at least 30 minutes.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial or send centrifuged serum gel tube.

4. For red top tubes, aliquot serum into a plastic vial within 10 minutes of centrifugation.

Specimen Minimum Volume: Serum 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	8 days

	Frozen	90 days

CPT Code Information:

0375U

OVMU
82825

Ovomucoid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

OXI
82679

Ox-Eye Daisy, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

DOXA1 Oxalate Analysis, Hemodialysate
606473

Specimen Requirements:

Specimen Type: Dialysate fluid

Patient Preparation: Patient should avoid taking vitamin C supplements for 24 hours prior to dialysis

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions: Adjust the pH of the specimen to 2.5 to 3.0 with 6M Hydrochloric Acid

Additional Information: Nonacidified frozen hemodialysate delivered to the laboratory within 3 days from collection will be accepted and the following comment will be added to the result: **In nonacidified hemodialysate stored frozen, oxalate values may increase spontaneously**.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Dialysate Fluid	Frozen	14 days

CPT Code Information:

83945

OXU
606737

Oxalate, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: For 24 hours before, as well as during the collection process, patient should not take large doses (>2 g orally/24 hours) of vitamin C.

Supplies:

-Diazolidinyl Urea (Germall) 5.0 mL (T822)

-Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube: Plastic tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea (Germall) as a preservative at start of collection or refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Mix container thoroughly and aliquot urine into plastic vial.

4. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. **Do not** attempt to adjust pH, as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

POXA1
606472

Oxalate, Plasma

Specimen Requirements:

Any client who has never collected a specimen for this test should call 800-533-1710 or 507-266-5700 and ask for the Clinical Specialty Laboratory for more detailed instructions.

Patient Preparation:

1. Fasting: 12 hours, preferred but not required

2. For 24 hours before specimen collection, the patient should not take vitamin C supplements.

Specimen Type: Acidified plasma

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Place specimen on wet ice immediately.

2. Within 1 hour of collection, centrifuge for 10 minutes at 3500 rpm. Use of a 4 degrees C refrigerated centrifuge is optimal but not required.

3. Aliquot plasma into a plastic vial.

4. Adjust the pH of the plasma specimen to a pH of 2.3-2.7 with approximately 10 mcL concentrated (12M) hydrochloric acid (or 20 mcL of 6M HCl) per 1 mL plasma.

Additional Information: Nonacidified specimens can be accepted if the heparinized plasma is properly frozen. A disclaimer will be added in nonacidified plasma, "Sample was received nonacidified and frozen. In nonacidified samples oxalate values may increase spontaneously." Although there can be a small absolute increase of plasma oxalate in nonacidified specimens, this will not change the clinical interpretation.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Frozen	30 days

CPT Code Information:

83945

ROXUR Oxalate, Random, Urine
606747

Specimen Requirements:

Patient Preparation: Avoid taking large doses (>2 g orally/24 hours) of vitamin C prior to specimen collection.

Supplies: Urine Tubes, 10 mL tube (T068)

Container/Tube: 10-mL plastic tube or a clean, plastic container with no metal cap

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 may indicate bacterial contamination, and testing will be cancelled. **Do not** attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

82570

OXCO1 Oxalate, Random, Urine (Non-Orderable)
606748

Specimen Requirements:

Only orderable as part of a profile. For more information see ROXUR / Oxalate, Random, Urine.

Patient Preparation: Avoid taking large doses (>2 g orally/24 hours) of vitamin C prior to specimen collection.

Supplies: Urine Tubes, 10 mL tube (T068)

Container/Tube: 10-mL plastic tube or a clean, plastic container with no metal cap

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 may indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

83945

RAT11 Oxalate/Creatinine Ratio, Urine (Non-Orderable)

606751

Specimen Requirements:

Only orderable as part of a profile. For more information see ROXUR / Oxalate, Random, Urine.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

OXAZS Oxazepam, Serum

623016

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

OMHC 81030 Oxcarbazepine Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST is **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80183

OXYDS
623456

Oxycodone and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYSX
61727

Oxycodone Screen, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 20 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

OXYSU
62623

Oxycodone Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5 mL tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

OXYCX 61728 Oxycodone with Metabolite Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain of custody kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 5 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain of custody testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYCU 62616 Oxycodone with Metabolite Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYMS 623460 Oxymorphone and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYMU 62622 Oxymorphone Confirmation, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80365 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

OXYWB 113429 Oxysterols, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin) or yellow top (ACD B)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	72 hours
	Ambient	48 hours

CPT Code Information:

82542

OXYBS 63147 Oxysterols, Blood Spot

Specimen Requirements:

Supplies:

-Card-Blood Spot Collection (Filter Paper) (T493)

-Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Blood Spot Collection Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 filter paper, PerkinElmer 226 filter paper,

Munktell filter paper, Postmortem Screening Card or collected with EDTA, sodium heparin, lithium heparin, or ACD B-containing devices

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Let blood dry completely on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

2. At least 1 spot should be complete, (ie, unpunched).

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	10 days	FILTER PAPER
	Frozen	59 days	FILTER PAPER
	Ambient	10 days	FILTER PAPER

CPT Code Information:

82542

OXNP 62988 Oxysterols, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin or lithium heparin), yellow top (ACD B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions:

1. Centrifuge at 4 degrees C.

2. Aliquot plasma into plastic vial. **Do not disturb the buffy coat layer**.

3. Send frozen.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	65 days

CPT Code Information:

82542

OYST
82883

Oyster, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

CCPQ 81185 P/Q-Type Calcium Channel Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MGLE / Myasthenia Gravis/Lambert-Eaton Myasthenic Syndrome Evaluation, Serum

PNEFS / Neuroimmunology Antibody Follow-Up, Serum

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86596

P16 70524 p16 (INK4a/CDKN2A) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

P40NA 70526 p40 + Napsin A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

P40
70527

p40 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P504S
610458

P504S Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P53
70528

p53 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections

4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P57I 70529 p57 (KIP2/CDKN1C) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

P62 70629 p62 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC,If additional IHC

P63 70530 p63 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

SQUI
82821

Pacific Squid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PANOX 62737 Pain Clinic Survey 10, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 30 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

PN10X 62911 Pain Clinic Survey 10, Chain of Custody, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PANOX / Pain Clinic Survey 10, Chain of Custody, Random Urine.

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and documentation is required.

Specimen Volume: 30 mL

Collection Instructions: Collect a random specimen without preservative in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 30 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 20 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80307

FPALI
75392

Paliperidone, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Serum

Collection Container/Tube: Red

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Draw blood in a plain, red-top tube(s). (Serum gel tube is not acceptable.) Spin down and send 1 mL of serum refrigerate in a preservative-free plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	

CPT Code Information:

80342

FPAN2 Pancreastatin, Plasma
75907

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 hours, recommended; water, but no other liquid, may be taken as needed

2. Patient should not be on any medications that may influence insulin levels, if possible, for at least 48 hours prior to collection.

Collection Container/Tube: Z tube (T701)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Draw 10 mL of blood in a pre-chilled Z-tube.

2. Immediately after collection, centrifuge in a refrigerated centrifuge and aliquot 3 mL of plasma into a plastic vial.

3. Freeze immediately.

4. Send frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Z-Tube Plasma	Frozen	60 days

CPT Code Information:

83519

ELASF
609492

Pancreatic Elastase, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a fresh random fecal specimen, no preservatives.

2. If specimen is sent refrigerate, send immediately after collection.

3. If specimen cannot be sent immediately, freeze specimen and send frozen.

Additional Information:

1. **Separate specimens must be submitted when multiple tests are ordered**, with the exception of CALPR / Calprotectin, Feces. **If only a single specimen is collected, it must be split prior to transport**.

2. Testing cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	28 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

82653

HPP
8014

Pancreatic Polypeptide, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, required

2. The patient **should not** have recently received radioisotopes, either therapeutically or

diagnostically, due to potential assay interference. See Ordering Guidance.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Plasma

Collection Instructions:

1. Place specimen on wet ice and keep cold at all times following collection.

2. Centrifuge (refrigerated centrifuge is not required) and aliquot plasma into a plastic vial. Freeze immediately.

Specimen Minimum Volume: Plasma: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

83519

WESPR 621326 Panel to Whole Exome Sequencing Reflex Test, Varies

Specimen Requirements:

For most patients, a new specimen submission will not be required. Testing can be performed using stored DNA from the original whole exome sequencing test. To order testing on the stored specimen, see Additional Testing Requirements.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report.

Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated

to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81417-Patient only

81417, 81416-Patient and one family member comparator sample (duo) (as appropriate)

81417, 81416 x 2-Patient and two family member comparator samples (trio or non-traditional trio) (as appropriate)

81417, 81416 x 3-Patient and three family member comparator samples (quad) (as appropriate)

PAPY
82356

Papaya, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PAPR
82810

Paprika, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PFIB
601950

Parafibromin, Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PNPAB
61881

Paraneoplastic Pemphigus Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86255

PVLE
607409

Paraneoplastic Vision Loss Evaluation, Serum

Specimen Requirements:

Patient Preparation:

1. For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication.

2. This test should not be requested for patients who have recently received radioisotopes, therapeutically or diagnostically, because of potential assay interference. The specific waiting period before specimen collection will depend on the isotope administered, the dose given, and the clearance rate in the individual patient. Specimens will be screened for radioactivity prior to analysis. Radioactive specimens received in the laboratory will be held 1 week and assayed if sufficiently decayed or canceled if radioactivity remains.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x1

84182 x1

84182 CRMWS (if appropriate)

86256 CRMTS (if appropriate)

PARID
9202

Parasite Identification, Varies

Specimen Requirements:

Specimen Type: Parasitic worms, insects, or mites

Container/Tube: Sterile container (10% formalin or 70% alcohol may be added if appropriate specimen type)

Specimen Volume: Entire specimen

Collection Instructions:

1. For scabies, submit skin scrapings on glass microscope slide. Cover with a clean slide and use a rubber band to hold the 2 slides together (do not tape). Place the slides in a clean, dry container for transport. If no slides are available at time of collection, the dry skin scraping can be submitted in a sterile container and will be placed on slide at time of examination. Skin scraping must be visible with the naked eye to be of a quantity sufficient for testing.

2. Submit whole worms and worm segments in 70% alcohol or formalin. Worms must be

clean of stool to be suitable for examination.

3. Submit arthropods (ticks, lice, nits, bed bugs, etc) in a clean, dry container. Do not wrap in tape, gauze, or tissue that might obscure them during examination.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87168-Arthropod (if appropriate)

87169-Parasite (if appropriate)

87210 - Arthropod or Parasite (if appropriate)

PTH
70544

Parathyroid Hormone (PTH) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PTHFN 61526 Parathyroid Hormone, Fine-Needle Aspiration Biopsy (FNAB)-Needle Wash

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. Needle wash specimens for analysis should be collected in conjunction with cytology specimens.

2. Have saline available prior to start of procedure. Saline is the only acceptable solution for needle washings.

3. After each fine-needle aspiration biopsy (FNAB) has been collected and the material in the needle has been expelled onto a slide for cytologic analysis, attach the used FNAB needle to an empty syringe.

4. Draw between 0.10 mL and 0.25 mL of saline up through the needle until the saline starts to fill the hub of the needle or end of the syringe.

5. Expel this fluid back through the needle into a separate plastic aliquot tube. This is the needle washing used for analysis.

6. Repeat steps 2 through 4 for each needle pass of the same biopsied site and empty into the same tube, accumulating a total of 0.5 mL to 1.5 mL of fluid to send to the laboratory. (If more than 1 site is biopsied, see Additional Information)

7. Inspect specimen for visible blood or tissue contamination:

a. If bloody, centrifuge specimen and transfer supernatant to a new plastic aliquot tube (5-mL standard tube) to send to laboratory. The supernatant, not the cellular material, is used for analysis.

b. If specimen is clear, centrifugation is not necessary.

8. Freeze within 2 to 4 hours of collection.

Additional Information:

1. If more than 1 site is biopsied, each washing material should be submitted on a separate tube and under a different order number.

2. A minimum of 0.5 mL is required for testing; however, the total collection volume should not exceed 1.5 mL. Sample volumes outside these parameters may be rejected.

3. Do not send saline control. This test has been validated to rule-out saline matrix effect.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fine Needle Wash	Frozen (preferred)	30 days
	Refrigerated	4 hours

CPT Code Information:

83970

PTH2 Parathyroid Hormone, Serum
28379

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, preferred but not required

2. **For 12 hours before specimen collection,** patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

83970

PTHRP
81774

Parathyroid Hormone-Related Peptide, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Ice-cooled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Plasma

Collection Instructions:

1. Centrifuge specimen in a refrigerated centrifuge or in chilled centrifuge cups.

2. Aliquot plasma into plastic vial and freeze.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

82397

PPAP
52964

Parental Sample Prep for Prenatal Microarray Testing, Blood

Specimen Requirements:

This test requires 2 blood specimens: 1 sodium heparin and 1 EDTA

Specimen Type: Whole blood

Container/Tube: Green top (sodium heparin) **and** lavender top (EDTA)

Specimen Volume: 3 mL EDTA **and** 4 mL sodium heparin

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimens in original tubes. **Do not aliquot.**

Specimen Minimum Volume: 2 mL EDTA and 2 mL sodium heparin whole blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

This test ID contains no charge and serves as a way to correlate proband parental specimens. If additional testing is warranted, the appropriate tests will be added.

PCAB
83728

Parietal Cell Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

PJUD
82877

Parietaria judaica, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

POFF
82549

Parietaria officinalis, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PLINK
62139

Paroxysmal Nocturnal Hemoglobinuria, PI-Linked Antigen, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 2.6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry, RBC x 1

88184-Flow cytometry, WBC x 1

88185-Flow cytometry, additional marker (each), RBC x 1

88185-Flow cytometry, additional marker (each), WBC x 6

88188-Flow Cytometry Interpretation, 9-15 Markers x 1

FPRTF 57967 Parrot Australian (Budgerigar) Feathers IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPARG 57686 Parsley IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PSLY
82765

Parsley, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PARVS
48395

Parvovirus B19 Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747 x 2

PARVG 48320 Parvovirus B19 Antibodies, IgG, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747

PARVM 48321 Parvovirus B19 Antibody, IgM, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86747

PARVN 48322 Parvovirus B19 Antibody, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PARVS / Parvovirus B19 Antibodies, IgG and IgM, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

PARVP
86337

Parvovirus B19, Molecular Detection, PCR, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PARVO
83151

Parvovirus B19, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Do not centrifuge.

2. Label specimen as amniotic fluid.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Do not centrifuge.

2. Label specimen as spinal fluid.

Specimen Type: Synovial fluid

Container/Tube: Sterile vial or lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as synovial fluid.

Alternate:

Specimen Type: Bone marrow

Container/Tube: Sterile container or lavender top (EDTA)

Specimen Volume: 0.5 mL

Collection Instructions: Label specimen as bone marrow.

Specimen Minimum Volume: Amniotic Fluid, Bone Marrow, and Spinal Fluid 0.3 mL; Synovial fluid 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PARVI
70532

Parvovirus Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PFRUT
82355

Passion Fruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CONFL
620379

Pathology Consultation

Specimen Requirements:

Additional Information:

All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Specimen type and anatomic site

Supplies: MCF Pathology Packaging Kit (T994)

Specimen Type: Tissue

Container/Tube: Formalin fixed paraffin-embedded blocks, stained slides and/or unstained slides

Note: Submit only pertinent hematoxylin and eosin (H and E) and all special stains relevant to the diagnostic questions for the case. Include positively charged unstained slides and/or a formalin-fixed, paraffin-embedded tissue block if it is anticipated that additional stains or ancillary testing may be necessary.

Bone Marrow Case Specimen Requirements

For information on collecting specimens, see Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline.

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow aspirate clot embedded in paraffin block

Collection Instructions: Process and embed clot in paraffin

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core embedded in paraffin block

Collection Instructions: Process and embed core in paraffin.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
MMLDRY	Ambient	

CPT Code Information:

88321 (if appropriate)

88323 (if appropriate)

88325 (if appropriate)

PATHC 70317 Pathology Consultation

Specimen Requirements:

Additional Information:

All specimens and additional paperwork must be labeled with:

-Two patient identifiers (first and last name, date of birth, or medical record number)

-Specimen type and anatomic site

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Tissue

Container/Tube: Formalin-fixed paraffin-embedded blocks, stained slides, and/or unstained slides

Note: Submit only pertinent hematoxylin and eosin (H and E) and all special stains relevant to the diagnostic questions for the case. Include positively charged unstained slides and/or a formalin-fixed, paraffin-embedded tissue block if it is anticipated that additional stains or ancillary testing may be necessary.

Bone Marrow Case Specimen Requirements

For information on collecting specimens, see Bone Marrow Core Biopsy, Clot, and Aspirate Collection Guideline.

Specimen Type: Bone marrow aspirate

Slides: 5

Preferred: Freshly prepared slides made at the time of specimen collection

Acceptable: Slides made from EDTA specimen, within 2 hours of collection

Submission Container/Tube: Plastic slide holder

Specimen Type: Bone marrow aspirate clot

Container/Tube: Bone marrow aspirate clot embedded in paraffin block

Collection Instructions: Process and embed clot in paraffin

Specimen Type: Bone marrow core biopsy

Container/Tube: Fixed biopsy core embedded in paraffin block

Collection Instructions: Process and embed core in paraffin.

Specimen Type: Blood

Slides: 2

Preferred: 2 Freshly prepared fingerstick slides

Acceptable: 2 Slides made from whole blood in EDTA, made within 8 hours of collection

Submission Container/Tube: Plastic slide holder

Transport Temperature:

Specimen Type	Temperature	Time
MMLDRY	Ambient	

CPT Code Information:

88321 (if appropriate)

88323 (if appropriate)

88325 (if appropriate)

PAX5 PAX-5 Immunostain, Technical Component Only
70533

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PAX2 PAX2 Immunostain, Technical Component Only
607795

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigered	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PAX8
70534

PAX8 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PDGF 58102 PDGFB (22q13), Dermatofibrosarcoma Protuberans/Giant Cell Fibroblastoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 2 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

FBEP
57935

Pea Black-Eyed/Cow Pea (Vigna sinensis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPGNG 57654 Pea Green IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPEAC 57666 Peach IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PECH
82816

Peach, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPNTG 57537 Peanut IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPNG4 57571 Peanut IgG4

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PEANT 64756 Peanut, IgE with Reflex to Peanut Components, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86003

PEAN 82888 Peanut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

86003

PEAR
82807

Pear, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPCFG 57688 Pecan Food IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PCANH 62600 Pecan Hickory, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

PEC
82880

Pecan-Food, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

AMLFP
622390

Pediatric Acute Myeloid Leukemia Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x22, 88275x11, 88291 x1-FISH Probe, Analysis, Interpretation; 11 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

PAS38 Pediatric Allergy Screen 3 to 8 Years, Serum

83346

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 6

PAS3 Pediatric Allergy Screen

83345

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

PAS8
83347

Pediatric Allergy Screen >8 Years, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

PCDES
605129

Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended prior to initiation of immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86341

86363

86053

86255 x 12

86256 AMPIS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86363 MOGTS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

86053 NMOTS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

PCDEC 605130 Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 13

86341

86053

86256 AMPIC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86256 NMDIC (if appropriate)

86053 NMOTC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

BALFP 622400 Pediatric B-Lymphoblastic Leukemia/Lymphoma Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x19, 88275 x9, 88291 x1-FISH Probe, Analysis, Interpretation; 9 probe sets

88271 x2, 88275 x1-FISH Probe, Analysis; each additional probe set (if appropriate)

TALFP 622420 Pediatric T-Lymphoblastic Leukemia/Lymphoma Panel, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x14, 88275x7, 88291x1- FISH Probe, Analysis, Interpretation; 7 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

PBPO
82660

Penicillin G, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENIV
82656

Penicillin V, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENL
82913

Penicillium chrysogenum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PENTS
8239

Pentobarbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before the next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FPBPG
57657

Pepper Bell/Paprika IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPBLG
57645

Pepper Black IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPCYE
57538

Pepper Cayenne (Capsicum frutescens) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days

	Frozen	28 days

CPT Code Information:

86003

FPEPI
75916

Pepsinogen I (PG I)

Specimen Requirements:

Patient preparation:

1. Fasting: 10 to 12 hours, recommended

2. Antacids or other medications affecting stomach acidity or gastrointestinal motility should be discontinued, if possible, for at least 48 hours prior to collection.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Centrifuge and aliquot 3 mL serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

83520

PERAM
609438

Perampanel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top (serum gel/SST is **not acceptable**)

Acceptable: None

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. For sustained-release formulations ONLY, draw blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

SAT
2503

Percent Saturation (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see SFEC / Iron and Total Iron-Binding Capacity, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

FOPE
57938

Perch Ocean IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SMPB
37406

Peripheral Blood Smear Review (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood Slide	Refrigerated		CARTRIDGE

CPT Code Information:

85060

PNBX
70598

Peripheral Nerve Pathology Consultation

Specimen Requirements:

Specimen Type: Nerve biopsy tissue, slides, or block

Supplies: Nerve Biopsy Collection Kit (T457)

Container/Tube: Nerve Biopsy Collection Kit

Collection Instructions: Prepare and transport specimen per Nerve Biopsy Specimen Preparation Instructions (T580).

Specimen Minimum Volume: 4.5 cm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Frozen	

CPT Code Information:

88305 (if appropriate)

88313 (if appropriate)

88321 (if appropriate)

88323 (if appropriate)

88323-26 (if appropriate)

88325 (if appropriate)

88362 (if appropriate)

88348 (if appropriate)

88342 (if appropriate)

88341 (if appropriate)

DMNES 621103 Peripheral Nervous System Demyelinating Neuropathy, Autoimmune Evaluation, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended to occur before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516 x5

83520

86255 x2

83520 x4 (if appropriate)

ACASM 83632 Pernicious Anemia Cascade, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required

2. For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

3. For 1 week before specimen collection, if medically feasible, patient **should not** take proton pump inhibitors (omeprazole, lansoprazole, dexlansoprazole, esomeprazole, pantoprazole, and rabeprazole).

4. For at least 2 weeks before specimen collection, patient **should not** take or receive drugs that interfere with gastrointestinal motility (eg, opioids).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL serum

Collection Instructions:

1. Centrifuge, divide specimen into 3 plastic vials:

Vial 1 (B12PA): 1 mL of serum

Vial 2 (PAMMA): 1.5 mL of serum

Vial 3 (PAGAS): 0.5 mL of serum

2. Band specimens together and send frozen.

Specimen Minimum Volume: Serum: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	24 hours

CPT Code Information:

82607

82941-if appropriate

83921-if appropriate

86340-if appropriate

PDGP
608013

Peroxisomal Disorder Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable,

specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24

hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

PNZN 9789 Perphenazine, Serum or Plasma

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Plasma

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Green-top (sodium heparin) (Plasma gel/PST are **not** acceptable)

Submission Container/Tube: Amber plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a green-top (sodium heparin) tube.

2. Centrifuge and aliquot 3 mL plasma into an amber vial to protect from light and refrigerate immediately.

3. Ship refrigerated.

Specimen Type: Serum

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Red top (Serum gel/SST are **no**t acceptable)

Submission Container/Tube: Amber Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in a plain red-top tube.

2. Centrifuge and aliquot 3 mL of serum into an amber vial to protect from light and refrigerate immediately.

3. Ship refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	180 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

80299

PERS 82353 Persimmon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STK1Z 614588 Peutz-Jeghers Syndrome, STK11, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole Blood 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

UPH24 pH, 24 Hour, Urine
606521

Specimen Requirements:

Supplies:

Diazolidinyl Urea (Germall), 5.0 mL (T822)

Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours

2. Add 5 mL of diazolidinyl urea as preservative at start of collection **or** refrigerate specimen during and after collection.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH >8 indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83986

UPHB
606522

pH, Body Fluid

Specimen Requirements:

Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	7 days
	Ambient	24 hours

CPT Code Information:

83986

FPHFL
57309

pH, Fecal

Specimen Requirements:

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a random unpreserved fecal specimen

2. Send frozen

Note: **Separate specimens must be submitted when multiple tests are ordered**.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

83986

PHU_ 606510 pH, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collections Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83986

FPHAS 57580 Phadiatop (Allergy Screen)

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	28 days

CPT Code Information:

86005

PHCGD 621575 Phagocytic Disorders and Chronic Granulomatous Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing, Tissue. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions.

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81223

81249

81479

81479 (if appropriate for government payers)

PCPMX 62740 Phencyclidine (PCP) Confirmation, Chain of Custody, Meconium

Specimen Requirements:

Supplies: Chain of Custody Meconium Kit (T653)

Container/Tube: **Chain of custody kit containing the specimen containers, seals, and documentation required**

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium specimen in the container provided. Seal and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Specimen that arrives with a broken seal does not meet the chain-of-custody requirements.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPMC 89069 Phencyclidine (PCP) Confirmation, Meconium

Specimen Requirements:

Supplies: Stool container. Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 1 g (approximately 1 teaspoon)

Collection Instructions: Collect entire random meconium (newborn's first bowel movements) specimen.

Specimen Minimum Volume: 0.3 g (approximately 1/4 teaspoon)

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPX 62739 Phencyclidine Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-Custody Kit containing the specimen containers, seals, and required documentation

Specimen Volume: 10 mL

Collection Instructions: Collect specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPU
80371

Phencyclidine Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container Tube: Plastic, 10 mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not** accepted for this test.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PCPS
624109

Phencyclidine Confirmation, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

G0480

83992 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FCDUA
75785

Phencyclidine, Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

PBR
37049

Phenobarbital, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80184

FPGT
91757

Phenosense Combination HIV Drug Resistance Assay

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood in either PPT (pearl top) or lavender-top (EDTA) tubes. Centrifuge specimen within six hours of collection. Remove plasma from cells immediately, and transfer specimen to a screw-capped, plastic vial. Freeze 3 mL of PPT plasma or EDTA plasma immediately, send specimen frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load.

2. Viral load collection date.

NOTE: 1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within 2 weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be cancelled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87900/Infectious agent drug susceptibility phenotype prediction

87901/Infectious agent genotype analysis by nucleic acid; reverse transcriptase and protease

87903/Infectious agent phenotype analysis by nucleic acid with drug resistance tissue culture analysis; first through 10 drugs tested

87904/x12 Each additional drug tested

FPFUZ Phenosense Entry HIV Drug Resistance Assay

91755

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood into **two** 5-mL PPT (pearl top) or EDTA (lavender top) tube. Centrifuge specimen within six hours of collection. Transfer plasma to one or more polypropylene screw-capped tube(s) and freeze. Freeze immediately. Send 3 mL plasma in a screw-cap vial frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load

2. Viral load collection date

NOTE:

1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be canceled.

Specimen Minimum Volume: 1.0 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87903

FPHIV 91756 Phenosense HIV Drug Resistance Replication Capacity

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volumes: 3 mL

Collection Instructions: Draw blood into **two** 5-mL PPT (pearl top) or EDTA (lavender top) tubes. Centrifuge specimen within 6 hours of collection. Transfer plasma to one or more polypropylene screw-capped tube(s) and freeze. Freeze immediately. Send 3 mL plasma in a screw-cap vial frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load.

2. Viral load collection date.

NOTE:

1. Intended for use only for patients with viral loads greater than or equal to 500 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be canceled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87903

87904 x12

PKUBS 65593 Phenylalanine and Tyrosine, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 90 days/Refrigerated 90 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate 6 days

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84030

84510

82542 (if appropriate for government payers)

PKU
8380

Phenylalanine and Tyrosine, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required; Infants should have specimen collected before next feeding

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Green top (lithium heparin), lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84030 Phenylalanine

84510 Tyrosine

82542 (if appropriate for government payers)

PKUSC
610508

Phenylalanine and Tyrosine, Self-Collect, Blood Spot

Specimen Requirements:

Container/Tube: Blood Spot Card

Specimen Volume: 2 Blood spots

Additional Information:

1. Order test each time the patient is to collect a dried blood specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address (**post office [PO] boxes are not acceptable delivery locations**), city, state abbreviation, zip code, country, and home phone number.

4. For each order, the Blood Spot Collection-Self Collect kit will be mailed directly to the patient for self-collection (**delivery to a PO box will not occur**).

5. For more information on how to collect blood spots, see the following:

-How to Collect a Dried Blood Spot Sample via fingerstick.

-Blood Spot Collection Instructions-Fingerstick

-Blood Spot Collection Instructions-Fingerstick-Spanish

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Refrigerated	90 days	FILTER PAPER

CPT Code Information:

0382U

PHEGP 608032 Phenylalanine Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test

(including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 2

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

PNYF
37052

Phenytoin, Free, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days

	Ambient	7 days

CPT Code Information:

80186

PNTFT 37051 Phenytoin, Total and Free, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

Phenytoin, total-80185

Phenytoin, free-80186

PNYG 37050 Phenytoin, Total and Phenobarbital Group, Serum

Specimen Requirements:

One serum specimen (0.5 mL of serum) may be sent if using a red top tube.

Serum for Phenytoin:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Serum for Phenobarbital:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL for 2 specimens; 0.25 mL for 1 serum red top

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

80184-Phenobarbital

80185-Phenytoin, total

PNYA
37048

Phenytoin, Total, Serum

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

80185

PHMA
82736

Phoma betae, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PTEN
614126

Phosphatase and Tensin Homolog (PTEN) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Preferred:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Acceptable: None

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PETH
617480

Phosphatidylethanol Confirmation, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Do not use alcohol to clean arm. Use alternative (such as Betadine, also known as Povidone Iodine) to cleanse site before specimen collection.

2. Draw blood in tubes smaller than 10 mL if possible.

3. Do not centrifuge.

4. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Frozen (preferred)	28 days
	Refrigerated	14 days

CPT Code Information:

G0480

80321 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

PSPTG
62578

Phosphatidylserine/Prothrombin Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86148

PT217 621635 Phospho-Tau 217, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic screw-top vial

Specimen Volume: 0.6 mL plasma

Collection Information: Centrifuge and aliquot plasma into plastic vial. **Do not submit in original tube.**

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
EDTA Plasma	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

84393

PDETS 620070 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PDETC 620069 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PDEIS 620068 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

PDEIC 620067 Phosphodiesterase 10A (PDE10A) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

PETNU 623429 Phosphoethanolamine, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Clean plastic urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82131

PFK1
607456

Phosphofructokinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82657

PFKC 608422 Phosphofructokinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82657

PGK1 607457 Phosphoglycerate Kinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA) or yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

PGKC 608423 Phosphoglycerate Kinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

PLAIF 70592 Phospholipase A2 Receptor (PLA2R), Renal Biopsy

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unstained slides (unfixed)

Source: Kidney tissue

Slides: 2 Slides

Collection Instructions: 2 frozen tissue unstained positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick, **centered on the slide,** and

submitted on dry ice.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Acceptable:

Specimen Type: Unfixed tissue block (O.C.T)

Source: Kidney tissue

Specimen Volume: Entire specimen

Collection Instructions:

1. Embed in O.C.T. compound.

2. Freeze specimen and ship on dry ice.

Acceptable:

Specimen Type: Wet tissue

Source: Kidney tissue

Supplies: Renal Biopsy Kit (T231)

Container/Tube: Renal Biopsy Kit, Zeus/Michel's

Specimen Volume: Entire specimen

Collection Instructions: Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

Additional Information: If standard immunoglobulin and complement immunofluorescence have already been performed, submit the residual frozen tissue (must contain glomeruli) on dry ice.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Frozen	

CPT Code Information:

88346-primary IF

EURO 606475 Phospholipase A2 Receptor, Enzyme Linked Immunosorbent Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PMND1 / Primary Membranous Nephropathy Diagnostic Cascade, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial within 2 hours of collection

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

PLA2I 607367 Phospholipase A2 Receptor, Immunofluorescence, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

86255

PLA2M
607366

Phospholipase A2 Receptor, Monitoring, Enzyme-Linked Immunosorbent Assay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

ACLIP
86179

Phospholipid (Cardiolipin) Antibodies, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

CLPMG
82976

Phospholipid (Cardiolipin) Antibodies, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147 x 2

GCLIP 80993 Phospholipid (Cardiolipin) Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

MCLIP 81900 Phospholipid (Cardiolipin) Antibodies, IgM, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86147

PMMIL 89656 Phosphomannomutase and Phosphomannose Isomerase, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 6 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood ACD	Refrigerated (preferred)	6 days	YELLOW TOP/ACD
	Ambient	6 days	YELLOW TOP/ACD

CPT Code Information:

82657

PHOS
8408

Phosphorus (Inorganic), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	365 days
	Refrigerated	7 days

CPT Code Information:

84100

PHOSS
622796

Phosphorus (Inorganic), Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

-RTRP2 / Tubular Reabsorption of Phosphorus, Random Urine and Serum

-TRPP / Tubular Reabsorption of Phosphorus, Serum/Plasma/Urine

Patient Preparation: Fasting 12 hours, required

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

POU
610832

Phosphorus, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84105

RPHOC
610829

Phosphorus, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84105

STAT3 622778 Phosphorylated STAT3 at Tyrosine 705 (pSTAT3-Y705), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PHTDP 71482 Phosphorylated TDP43 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PAHD
82786

Phthalic Anhydride, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CVSPC
72163

Physician Interpretation Conventional, Varies

Specimen Requirements:

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		SLIDE
	Refrigerated		SLIDE

CPT Code Information:

88141

TPSPC
72162

Physician Interpretation Screen, Varies

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		THIN PREP
	Refrigerated		THIN PREP

CPT Code Information:

88141

TPDPC
72129

Physician Interpretation, Diagnostic, Varies

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)		THIN PREP
	Refrigerated		THIN PREP

CPT Code Information:

88141

PIGE
82781

Pig Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPIGF
75555

Pigeon Feathers IgG

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PIGF
82145

Pigeon Feathers, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPDD
75548

Pigeon/Dove Droppings Gel Diffusion

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red or SST

Specimen Volume: 0.5 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86331

PINE
82381

Pine Nut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPINP
75410

Pine Ponderosa IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FPIAP
57670

Pineapple IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PNAP
82815

Pineapple, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PINW
9204

Pinworm Exam, Perianal

Specimen Requirements:

Specimen Type: Perianal

Container/Tube:

Preferred: ParaClick Pinworm Paddle

Acceptable: SWUBE disposable paddle or similar method of collection

Specimen Volume: Entire specimen

Collection Instructions: See Pinworm Collection Instructions

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

87172

PIPU
81248

Pipecolic Acid, Random, Urine

Specimen Requirements:

Supplies: Urine tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	94 days
	Refrigerated	14 days

CPT Code Information:

82542

PIPA
81326

Pipecolic Acid, Serum

Specimen Requirements:

Patient Preparation:

Fasting 12 hours, required; Infants and small children should have specimen collected before next feeding/meal

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	94 days
	Refrigerated	14 days

CPT Code Information:

82542

PISTA
82808

Pistachio, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PIT1
72124

PIT-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PKLRZ
610058

PKLR Full Gene Analysis, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

PLAP
70539

Placental Alkaline Phosphatase (PLAP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PLAI
82837

Plaice, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBLI
9302

Plasma Cell Assessment, Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Specimen Volume: 10 mL

Collection Instructions:

1. **Do not centrifuge.**

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry, cell surface, cytoplasmic

88185 x 5-Each additional marker

88187-Flow cytometry, interpretation; 2 to 8 markers

PCPRO 61654 Plasma Cell DNA Content and Proliferation, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 4 mL

Specimen Stability Information: 3 days

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers (added as FCINT)

CSPMM 618626 Plasma Cell Myeloma Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see NGPCM / MayoComplete Plasma Cell Myeloma, Next-Generation Sequencing, Varies.

Container/Tube: Lavender or pink top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Minimum plasma cell percentage is 1%.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

4. Label specimen as bone marrow.

5. **Fresh specimen is required for this test**, as testing is performed on sorted cells.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

PLASF 35293 Plasma Cell Proliferative Disorder, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions: Submit 10 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 7 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PCPDS 606079 Plasma Cell Proliferative Disorder, High Risk with Reflex Probes, Diagnostic FISH Evaluation, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88271 x 2, 88274, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88274-FISH Probe, Analysis; each additional probe set (if appropriate)

CSPCF 607625 Plasma Cell Proliferative Disorder, Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see PCPDS / Plasma Cell Proliferative Disorder, High Risk with Reflex Probes, Diagnostic FISH Evaluation, Bone Marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

PLHBB 9096 Plasma Free Hemoglobin, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

2. Results could be falsely elevated due to artifactual red blood cell lysis if not centrifuged within 2 hours of collection.

Specimen Minimum Volume: Plasma: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	20 days
	Frozen	30 days
	Ambient	4 days

CPT Code Information:

83051

PGRBC
609675

Plasmalogens, Blood

Specimen Requirements:

Patient Preparation: Specimen must be collected either prior to or 6 weeks after a blood transfusion.

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), yellow top (ACD solution A or B)

Specimen Volume: 5 mL

Collection Instructions: Send specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Ambient	14 days

CPT Code Information:

82542

PGDBS 609664 Plasmalogens, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Patient Preparation: Specimen must be collected either prior to or 6 weeks after a blood transfusion.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood spot collection card

Acceptable: Local newborn screening card, PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, or blood collected in tubes containing ACD, EDTA, or heparin spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 90 days/Refrigerated 90 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), yellow top (ACD solution A or B)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Stability Information: Refrigerate (preferred) 14 days/Ambient 11 days

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82542

PAI1
86083

Plasminogen Activator Inhibitor Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, at-40 degrees C or below.

Specimen Stability Information: Frozen 2 years

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85415

PSGN 9079 Plasminogen Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and

Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85420

PLABN 35794 Platelet Antibody Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Additional Information: For testing neonates, draw blood **from the mother** of the neonate. Then follow the Collection Instructions prior to sending.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	365 days
	Refrigerated	48 hours

CPT Code Information:

86022

GNPLT
619285

Platelet Disorders, Comprehensive Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred) 24 hours/Ambient 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) 24 hours/Refrigerated 24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Cultured fibroblasts/skin biopsy: see Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

GNPFD 619355 Platelet Function Defect Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

Platelet Neutralization Procedure, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Specimen Stability Information: Frozen 2 years

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85597

GNSPD 619327 Platelet Storage Pool Deficiency Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710..

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

PLAFL 64278 Platelet Surface Glycoprotein, Flow Cytometry, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Collection Container/Tube: ACD solution A or B

Specimen Volume: 6 mL

Pediatric Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: Adult: 1 mL; Pediatric: 200 mcL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each) X5

88187-Flow cytometry interpretation, 2 to 8 markers

PTEM 63682 Platelet Transmission Electron Microscopic Study, Whole Blood

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Collection Container/Tube:

Preferred: Yellow top (ACD, solution B)

Acceptable: Yellow top (ACD, solution A)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Ambient	72 hours

CPT Code Information:

85390

88348

FPLAT 75450 Platinum, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Draw blood in a plain, royal blue top tube(s). **(Serum gel tube is not acceptable.)** Spin down and send 1 mL of serum refrigerated in an acid washed (MCL Supply T619) or

trace metal-free plastic container.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	60 days
	Ambient	60 days
	Frozen	60 days

CPT Code Information:

83018

PLUM
82809

Plum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PMLR
84114

PML::RARA Quantitative, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top **(**EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Blood: 8 mL; Bone Marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA

	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81315-PML/RAR-alpha (t(15;17)), (PML-RARA regulated adaptor molecule 1) (eg, promyelocytic leukemia) translocation analysis; all breakpoints (eg, intron 3, intron 6 and variable in exon 6), qualitative or quantitative

PMPDD PMP22 Gene, Large Deletion/Duplication Analysis, Varies

66569

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81324

PMS2 70540 PMS2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FPA15 Pneumococcal Antibody Panel - PCV15 (15-serotype)
75952

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Allow blood to clot for 30 to 60 minutes.

2. Centrifuge at ambient temperature, 1000 x g for 15 minutes.

3. Aliquot 1 mL of serum into a plastic vial.

4. Send refrigerated.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	42 days
	Ambient	42 days
	Frozen	

CPT Code Information:

86581

PNRP 81698 Pneumocystis jiroveci, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Pneumocystis* species DNA is unlikely.

Preferred Specimens: Pleural, respiratory (eg, bronchoalveolar lavage [BAL], bronchial washing, sputum), or fresh tissue

Acceptable Specimens: If no fresh specimen is available, digested respiratory specimens treated with N-acetyl-L-cysteine-sodium hydroxide (NALC/NaOH) are acceptable (eg, BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion)

Submit only 1 of the following specimens:

Preferred

Specimen Type: Body fluid

Sources: Pleural

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Only fresh, non-NALC/NaOH-digested body fluid is acceptable.

Specimen Type: Respiratory

Sources: BAL, bronchial washing, tracheal secretions, or sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL if only PCR ordered or 3 mL if PCR ordered with smear and culture

Specimen Type: Tissue

Sources: Respiratory

Container/Tube: Sterile container

Specimen Volume: 5-10 mm

Collection Instructions:

1. Submit fresh tissue.

2. Keep tissue moist with sterile water or sterile saline

Acceptable

Specimen Type: NALC/NaOH-digested respiratory specimens

Sources: BAL, bronchial washing, respiratory fluid, sputum, or tracheal secretion

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

1. Submit digested specimen treated with NALC/NaOH.

2. Clearly indicate on container and order form that specimen is a digested specimen.

Specimen Minimum Volume: Body fluid or nondigested respiratory specimen: 0.5 mL; Fresh tissue: 5 mm; NALC-NaOH-digested specimen: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87594

D240 70416 Podoplanin (D2-D40) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

POLET 619676 POLE Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81407

FPOLO 75165 Poliovirus (Types 1, 3) Antibodies, Neutralization

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Ship refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	5 days

CPT Code Information:

86382 x 2

FPOLE 57942 Pollock White (Pollachius virens) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PVJAK 65116 Polycythemia Vera, JAK2 V617F with Reflex to JAK2 Exon 12-15, Sequencing for Erythrocytosis, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	5 days

CPT Code Information:

81270-JAK2 V617

0027U (if appropriate)

TALDO
61843

Polyols, Quantitative, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

FPOM
57918

Pomegranate (Punica granatum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PD2T 65296 Pompe Disease Second-Tier Newborn Screening, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA and then spotted and dried on filter paper.

Specimen Volume: 3 Blood spots

Collection Instructions:

1. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

83789

PDBS
602280

Pompe Disease, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA, and dried on filter paper

Specimen Volume: 3 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry completely on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	28 days

CPT Code Information:

83789

GAAN 622582 Pompe Disease, GAA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. Do not aliquot.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a

minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen can only tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FPOPW 57557 Poplar White (Populus alba) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

POPSD
82632

Poppy Seed, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL; For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FPORG
57627

Pork IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FPRK4 Pork IgG4

57564

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

PORK
82700

Pork, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBALP
64661

Porphobilinogen and Aminolevulinic Acid, Plasma

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for at least 24 hours prior to specimen collection.

Supplies: Amber Frosted Tube, 5mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Green top (lithium heparin), lavender top (EDTA), yellow top (ACD A or B)

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. It is recommended that specimen collection occur during the acute phase. Porphobilinogen and aminolevulinic acid may be normal when the patient is not exhibiting symptoms.

2. Centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma	Frozen (preferred)	21 days	LIGHT PROTECTED
	Refrigerated	7 days	LIGHT PROTECTED

CPT Code Information:

82542

82135

PBGDW 31894 Porphobilinogen Deaminase, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol induces porphobilinogen deaminase activity, which may lead to a false-normal result.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days
	Ambient	48 hours

CPT Code Information:

82657

PBGD_ 88925 Porphobilinogen Deaminase, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol. This is essential as ethanol induces porphobilinogen deaminase activity, which may lead to a false-normal result.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or green top (lithium heparin)

Specimen Volume: 4 mL; full tube

Collection Instructions: Refrigerate specimen as soon as possible.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	8 days
	Ambient	7 days

CPT Code Information:

82657

PBGU
82068

Porphobilinogen, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for at least 24 hours before specimen collection.

Supplies: Urine Container-Amber, 60 mL (T596)

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative necessary but pH must be above 5.0. If pH is below 5.0, specimen will be rejected.

3. Specimens should be protected from light and frozen immediately following collection.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen (preferred)	7 days	LIGHT PROTECTED
	Refrigerated	7 days	LIGHT PROTECTED

CPT Code Information:

84110

PCGP
608023

Porphyria Comprehensive Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710**.**

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent Cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 2

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

PEWE
31893

Porphyrins Evaluation, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84311-Spectrophotometry, analyte not elsewhere specified

82542-Chromatography (if appropriate)

PEE
88886

Porphyrins Evaluation, Whole Blood

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), dark blue top (metal free heparin), royal blue top (metal free EDTA), green top (lithium heparin)

Specimen Volume: 4 mL

Collection Instructions: Immediately place specimen on wet ice.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

84311

82542-if appropriate

FQPPS
81652

Porphyrins, Feces

Specimen Requirements:

Patient Preparation:

1. For 3 days before, as well as during, specimen collection, patient **must not eat red meat or take any aspirin-containing medications.**

2. For 24 hours before, as well as during, specimen collection, patient **must not use** barium, laxatives, or enemas.

Supplies: Stool Containers - 24, 48, 72 Hour Kit (T291)

Container/Tube: Stool containers

Specimen Volume: Entire collection (48, 72, or 96 hours). 24-Hour collection is adequate if the collection volume is at least 100 g

Collection Instructions:

1. Collect all stool specimens within a 24, 48, 72, or 96-hour timeframe.

2. Do not add preservative.

3. Send entire collection.

Additional Information: Specimens smaller than 100 g may not provide accurate results.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

84126

PQNU 8562 Porphyrins, Quantitative, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: For the 24 hours before, as well as during, specimen collection, patient **must not** consume any alcohol.

Supplies:

-Urine Container - Amber, 60-mL (T596)

-Sodium Carbonate, 5 gram (T272)

Container/Tube: 60-mL Amber urine container

Specimen Volume: 20 to 50 mL

Collection Instructions:

1. Add 5 g of sodium carbonate as preservative at start of collection. This preservative is intended to achieve a pH above 7. **Do not substitute sodium bicarbonate for sodium carbonate.**

2. Collect a 24-hour urine specimen.

3. The container should be refrigerated and protected from light as much as possible during collection.

4. Record volume and duration.

5. Aliquot 20 to 50 mL into an amber container and freeze.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen	7 days	LIGHT PROTECTED

CPT Code Information:

84110-Porphobilinogen, quantitative

84120-Porphyrins, quantitation and fractionation

PQNRU 60597 Porphyrins, Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: For at least 24 hours before specimen collection, patient **must not** consume any alcohol.

Supplies: Urine Container - Amber, 60 mL (T596)

Container/Tube: 60-mL Amber urine container

Specimen Volume: 20 to 50 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative3. Specimens should be protected from light and frozen immediately following collection.

Specimen Minimum Volume: 15 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Urine	Frozen	72 hours	LIGHT PROTECTED

CPT Code Information:

84110-Porphobilinogen, quantitative

84120-Porphyrins, quantitation and fractionation

82135- ALA Delta Random Urine

PTP 8731 Porphyrins, Total, Plasma

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA)

Submission Container/Tube: Amber vial

Specimen Volume: 3 mL Plasma

Collection Instructions: Centrifuge specimen and aliquot plasma into an amber vial.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma	Frozen (preferred)	14 days	LIGHT PROTECTED
	Refrigerated	14 days	LIGHT PROTECTED

CPT Code Information:

84311-Porphyrins, total

82542-Porphyrins, fractionation (if appropriate)

FPPOS 75889 Posaconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen:

1. Fungus Test Lab Request Form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999 - Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

POSA
89591

Posaconazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge, and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80187

POSV
9205

Post Vasectomy Check, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Container/Tube: Semen Analysis Kit (T178)

Specimen Volume: Total ejaculate

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89321

PMAOG
620568

Postmortem Aortopathy Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81410

PMARG 620582 Postmortem Arrhythmia Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PMCAG 620596 Postmortem Cardiomyopathy and Arrhythmia Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PMCMG 620610 Postmortem Cardiomyopathy Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81439

PCMSP 620638 Postmortem Inherited Congenital Myasthenia Syndrome Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block.

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81443

PMHLH 620624 Postmortem Primary Hemophagocytic Lymphohistiocytosis (HLH) Gene Panel, Tissue

Specimen Requirements:

Specimen Type: Tissue (FFPE)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block

Additional Information: Testing will be attempted on blocks of any age but may be canceled if adequate DNA concentration cannot be obtained.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81443

PMSBB 81931 Postmortem Screening, Bile and Blood Spot

Specimen Requirements:

Both bile and blood spots are required.

Supplies: Card-Postmortem Screening (Filter Paper) (T525)

Container/Tube:

Preferred: Card-Postmortem Screening Card (Filter Paper)

Acceptable: Whatman Protein Saver 903 paper or local newborn screening card

Specimen Volume: Properly completed screening card

Collection Instructions:

1. Collect blood in a heparin-containing tube and drop 25 mcL of blood onto the 2 circles labeled Blood.

2. Collect bile by direct puncture of the gallbladder and drop 25 mcL of bile onto the 2 circles labeled Bile.

3. Allow to dry at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Fill out information on page 2 of collection card.

5. Do not expose specimen to heat or direct sunlight.

6. Do not stack wet specimens.

7. Keep specimen dry.

Specimen Minimum Volume: Bile spot: 1; Blood spot: 1

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)		FILTER PAPER
	Frozen		FILTER PAPER
	Refrigerated		FILTER PAPER

CPT Code Information:

83789

KUR 614060 Potassium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-Hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-Hour collection.

Additional Information:

See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84133

RKUR
610696

Potassium, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84133

KS
602352

Potassium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	48 hours

CPT Code Information:

84132

FPTWG
57539

Potato White IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

POU2F
618483

POU2F3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FMPG
57931

Poultry and Meat Panel IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001 x 7

POWV
616098

Powassan Virus, IgM, Enzyme-Linked Immunosorbent Assay, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86790

POWVB 617484 Powassan Virus, RNA, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVU 617487 Powassan Virus, RNA, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect random urine in a sterile container.

2. Label specimen as urine.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVS 617486 Powassan Virus, RNA, Molecular Detection, PCR, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions:

1. Within 2 hours of collection centrifuge and aliquot serum into a sterile container.

2. Serum specimens not aliquoted from the serum gel collection tube into a sterile container will be rejected.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

POWVC 617485 Powassan Virus, RNA, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Send specimen from collection vial 2.

2. Do not centrifuge or heat inactivate.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PWAS 35535 Prader-Willi/Angelman Syndrome, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks is required to culture amniotic fluid before genetic testing can occur.

3. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

3. **All prenatal specimens must be accompanied by a maternal blood specimen**; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81331-*SNRPN/UBE3A*, (small nuclear ribonucleoprotein polypeptide Nand ubiquitin protein ligase E3A) (eg, Prader-Willi syndrome and/or Angelman syndrome), methylation analysis

88235-Tissue culture for amniotic fluid (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Comparative analysis using Short Tandem Repeat (STR) markers; patient and comparative specimen (eg, pre-transplant recipient and donor germline testing, post-transplant non-hematopoietic recipient germline [eg, buccal swab or other germline tissue sample] and donor testing, twin zygosity testing or maternal cell contamination of fetal cells (if appropriate)

PRAME PRAME Immunostain, Technical Component Only

615794

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PALB Prealbumin, Serum

9005

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

84134

C2NAD
621651

PrecivityAD, Plasma

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding (T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: 2-mL screw cap micro tubes

Specimen Volume: 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Aliquot plasma into a 2 mL micro tube.

3. Freeze plasma (no longer than 2 hours after collection) at -20 degrees C or below.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0412U

C2AD2 PrecivityAD2, Plasma
621652

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding (T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: Two 2-mL screw cap micro tubes

Specimen Volume: 3 mL in 2 tubes, each containing 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Label two 2-mL screw-cap micro tubes.

3. Aliquot 1.5 mL of plasma into each labeled micro tube.

4. Freeze plasma (no longer than 2 hours after collection) at or below -20 degrees C.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0503U

AD2AR PrecivityAD2, Reflex to Apolipoprotein E, Plasma
621654

Specimen Requirements:

Supplies: Screw cap micro tube, 2 mL, PCR Performance Tested, Low protein-binding

(T983)

Collection Container/Tube: 10 mL Purple top (K EDTA)

Submission Container/Tube: Two 2-mL screw cap micro tubes

Specimen Volume: 3 mL in 2 tubes, each containing 1.5 mL

Collection Instructions:

1. Centrifuge within two hours of collection.

2. Label two 2-mL screw-cap micro tubes.

3. Aliquot 1.5 mL of plasma into each labeled micro tube.

4. Freeze plasma (no longer than 2 hours after collection) at or below -20 degrees C.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	

CPT Code Information:

0503U

PERA
621166

Preeclampsia sFlt-1/PlGF (Soluble fms-Like Tyrosine Kinase 1/ Placental Growth Factor) Ratio, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, the patient **should not** receive intravenous heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	24 hours

CPT Code Information:

0482U

PGN
65119

Pregabalin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

17PRN
88646

Pregnenolone and 17-Hydroxypregnenolone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	28 days

CPT Code Information:

84140-Pregnenolone

84143-17-Hydroxypregnenolone

PREGN
88645

Pregnenolone, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen and immediately aliquot serum into a plastic vial.

2. Freeze immediately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	28 days

CPT Code Information:

84140

PADF
35313

Prenatal Aneuploidy Detection, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mL.

4. If ordering with CHRAF / Chromosome Analysis, Amniotic Fluid, submit a minimum of 12 mL.

5. If ordering with both CMAP and CHRAF, then submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

Acceptable:

Specimen Type: Chorionic villi

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 30 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (Such as CVS Media (RPMI) and Small Dish).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

4. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic Fluid/Chorionic Villus Sampling, submit a minimum of 12 mg.

5. If ordering with CHRCV / Chromosome Analysis, Chorionic Villus Sampling, submit a minimum of 12 mg.

6. If ordering with both CMAP and CHRCV, then submit a minimum of 26 mg.

Specimen Minimum Volume: Amniotic fluid: 2 mL; Chorionic villi: 2 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		
	Ambient		

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

PHSP 5566 Prenatal Hepatitis Evaluation, Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.2 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot serum into a plastic vial and ship frozen (preferred).

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum SST	Frozen (preferred)	84 days	
	Refrigerated	6 days	

CPT Code Information:

87340

86803

G0472 (if appropriate for government payers)

87522 (if appropriate)

86707 (if appropriate)

87341 (if appropriate)

87350 (if appropriate)

PHEPU 620973 Previous Viral Hepatitis (Unknown Type), Serum

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient should **not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 2.6 mL

Collection Instructions:

1. Centrifuge blood collection tube per manufacturer's instructions (eg, centrifuge and aliquot within 2 hours of collection for BD Vacutainer tubes).

2. Aliquot 1.8 mL serum into a plastic vial and ship frozen (preferred).

Specimen Minimum Volume: 1.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	84 days
	Refrigerated	6 days

CPT Code Information:

86704

86706

86708

86803

87340

87341 (if appropriate)

87522 (if appropriate)

PBCPN 620737 Primary Biliary Cholangitis Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86039

83516 x2

86381

PCDGG 617421 Primary Ciliary Dyskinesia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

HLHGP 619830 Primary Hemophagocytic Lymphohistiocytosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical and Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory, or equivalent, and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When

applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

PMND1 609778 Primary Membranous Nephropathy Diagnostic Cascade, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

83520

86255 (x1 or x2, if appropriate)

PRMB 37053 Primidone and Phenobarbital, Serum

Specimen Requirements:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

PRIMD-80188

PBR-80184

PTRE
82784

Privet Tree, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PRKSG 617435 PRKAR1A Full Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination

studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

GAL2 606833 Probability of Hepatocellular Carcinoma, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see HCCGS / Hepatocellular Carcinoma Risk Panel with GALAD Score, Serum.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	5 days

PA 8683 Procainamide and N-Acetylprocainamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, plasma may sit on gel refrigerated or be aliquoted into a plastic vial.

3. For red top tubes, aliquot serum immediately into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

80192

PRCAL 602598 Procalcitonin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	48 hours
	Ambient	24 hours

CPT Code Information:

84145

PINP
61695

Procollagen I Intact N-Terminal, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

83519

LDAI
621645

Progentec aiSLE DX Lupus Disease Activity Index, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Yellow top (ACD solution A)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Centrifuge at 1500 x *g* for 20 minutes.

2. Aliquot plasma into plastic vial.

3. Freeze specimen within 2 hours of collection.

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	74 days

CPT Code Information:

0446U

LFRI
621644

Progentec aiSLE DX Lupus Flare Risk Index, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Yellow top (ACD solution A)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Centrifuge at 1500 x *g* for 20 minutes.

2. Aliquot plasma into plastic vial.

3. Freeze specimen within 2 hours of collection.

Note: Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	74 days

CPT Code Information:

0447U

PROG 70542 Progesterone Receptor (PR) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

PGSN 8141 Progesterone, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection and serum aliquoted into a plastic vial.

2. Red-top tubes should be centrifuged, and serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

84144

22C3 603762 Programmed Death-Ligand 1 (PD-L1) (22C3), Semi-

Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

288PD 609994 Programmed Death-Ligand 1 (PD-L1) (28-8), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1.One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

SP142 603769 Programmed Death-Ligand 1 (PD-L1) (SP142), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

SP263 603755 Programmed Death-Ligand 1 (PD-L1) (SP263), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This assay requires at least 100 viable tumor cells.

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

PINS
80908

Proinsulin, Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required; Infants younger than 2 years should fast a maximum of 6 hours

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Ice-cooled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after draw, place the whole blood on ice for at least 10 minutes.

2. Centrifuge at refrigerated temperature, aliquot plasma into a plastic vial, and freeze specimen.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	30 days

CPT Code Information:

84206

PRLI
70541

Prolactin (PRL) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PLPMA
35090

Prolactin, Pituitary Macroadenoma, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146

PRL
85670

Prolactin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146

APRI 603183 Prolonged Clot Time Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

APROL 603308 Prolonged Clot Time Profile, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 5 mL Platelet-poor plasma in five (5) 13-mm x 75-mm, 5-mL,

polypropylene, plastic vials , each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 5 separate 13-mm x 75-mm, 5-mL, polypropylene, plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: Four (4) 13-mm x 75-mm, 5-mL, polypropylene, plastic vials, each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85379-DIMER

85384-CLFIB

85390-26-APRI

85610-PTSC

85613-DRV1

85670-TTSC

85730-APTSC

85130-Chromogenic FVIII (if appropriate)

85130-Chromogenic FIX (if appropriate)

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85240-Factor VIII (if appropriate)

85250-Factor IX (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85335-Bethesda titer (if appropriate)

85335-Factor V inhibitor screen (if appropriate)

85335-Factor VIII inhibitor screen (if appropriate)

85335-Factor IX inhibitor screen (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirm (if appropriate)

85635-Reptilase time (if appropriate)

85732-APTT mix 1:1 (if appropriate)

FPHEG Promethazine (Phenergan)

90101

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FPIBD
75953

PROMETHEUS IBD Precis

Specimen Requirements:

Collection Container/Tube: Serum gel (red-top tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	14 days

	Frozen	30 days
	Ambient	7 days

CPT Code Information:

81599

PFN
80295

Propafenone, Serum

Specimen Requirements:

Patient Preparation: Specimens should only be collected after patient has been receiving propafenone for at least 3 days. Trough concentrations should be collected just before administration of the next dose.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FPROS
75902

Prostaglandin D2 (PG D2), Random Urine

Specimen Requirements:

Patient Preparation: Patient **should not** be on aspirin, indomethacin, or anti-

inflammatory medications, if possible, for at least 48 hours prior to collection of specimen.

Submission Container/Tube: Plastic vial

Specimen Volume: 10 mL

Collection Instructions:

1. Collect 10 mL random urine (24-hour urine collection is **not acceptable**)

2. No preservative.

2. Freeze immediately and send specimen frozen in a plastic vial.

3. Specimen must remain frozen prior to and during shipping.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	60 days

CPT Code Information:

84150

FPDPG 75900 Prostaglandin D2 (PGD2)

Specimen Requirements:

Patient preparation: Patient **should not** be on aspirin, indomethacin, or anti-inflammatory medications, if possible, for at least 48 hours prior to collection of specimen.

Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. Centrifuge as soon as possible, aliquot 3 mL of serum into a plastic vial.

2.Freeze specimen immediately after separation.

3. Send frozen.

Specimen Minimum Volume: Serum:1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	60 days

CPT Code Information:

84150

PHI11 Prostate Health Index Reflex, Serum
113000

Specimen Requirements:

Patient Preparation:

1. Specimens for testing should be collected prior to prostate manipulations such as digital rectal examination, prostatic massage, transrectal ultrasound, and prostatic biopsy.

2. A 6-week waiting period between needle biopsy and specimen collection is recommended.

3. Specimens should not be collected from patients receiving therapy with high biotin (vitamin B7) doses (ie, >5 mg/day) until at least 8 hours following the last biotin administration.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 3 hours of collection, centrifuge, aliquot serum into a plastic vial, and refrigerate. Freeze sample within 24 hours of collection and send frozen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	150 days

	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84153 - Total PSA

84154 - Free PSA (if appropriate)

86316 - [-2]ProPSA (if appropriate)

PSAIM 70543 Prostate Specific Antigen (PSA) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PTRIP
610456

Prostate Triple (P63/KRT/P504S), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

P2PHI
608462

Prostate-Specific Antigen (PSA) and [-2]ProPSA with Prostate Health Index (PHI), Serum

Specimen Requirements:

Patient Preparation:

1. Specimens for testing should be collected prior to prostate manipulations such as digital rectal examination (DRE), prostatic massage, transrectal ultrasound (TRUS), and prostatic biopsy.

2. A 6-week waiting period between needle biopsy and specimen collection is recommended.

3. Specimens should not be collected from patients receiving therapy with high biotin (vitamin B7) doses (ie, >5 mg/day) until at least 8 hours following the last biotin administration.

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 3 hours of collection, centrifuge, aliquot serum into a plastic vial, and refrigerate. Freeze sample within 24 hours of collection and send frozen.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	150 days
	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84153-Total PSA

84154-Free PSA

86316-[-2]ProPSA

PSA
9284

Prostate-Specific Antigen (PSA) Diagnostic, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions:

1. For serum gel tubes, centrifuge within 2 hours of collection.

2. For red top tubes, centrifuge and aliquot serum into a plastic vial within 2 hours of collection.

Additional Information: Free prostate-specific antigen (PSA) can only be added on within 72 hours of performing total PSA. Specimen must have been shipped frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	5 days

CPT Code Information:

84153

SPSA
82023

Prostate-Specific Antigen (PSA) Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

2. Serum gel tube must be centrifuged within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	5 days

CPT Code Information:

84153

G0103 (if appropriate)

PSAU
64061

Prostate-Specific Antigen (PSA) Ultrasensitive, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	180 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

84153

PSAFT
81944

Prostate-Specific Antigen (PSA), Total and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Within 3 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	72 hours

CPT Code Information:

84153

84154

PACPI 70531 Prostatic Acid Phosphatase (PACP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PACP 8019 Prostatic Acid Phosphatase, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

84066

CFX 9339 Protein C Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85303

PCAG 9127 Protein C Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Plasma

Specimen Stability Information: Frozen 2 years

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or ideally, at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	

CPT Code Information:

85302

GNPRC 619173 Protein C Deficiency, PROC Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

SFX
617792

Protein S Activity, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Patient Preparation: Patient must not be receiving Coumadin.

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze specimen immediately (no longer than 4 hours after collection) below -40 degrees C.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306

PSF
80338

Protein S Antigen, Free, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile, see PSTF / Protein S Antigen, Plasma.

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Coumadin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been